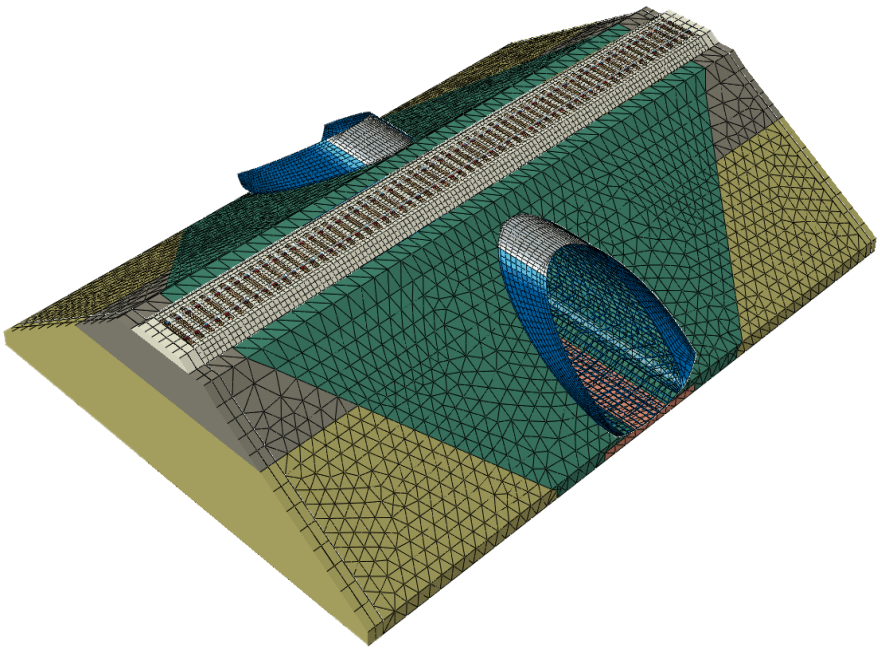


Dynamic Analysis of Soil-Steel Composite Bridges for High-Speed Railway Traffic

Case study of a bridge in Åre, Sweden using field measurements and FE-Analysis.

Degree Project in Civil and Architectural Engineering, KTH
Second cycle, 30 credits

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Abstract

Soil–steel composite bridges are commonly used for short-span railway applications due to their efficient construction and low material use. However, their dynamic behavior under high-speed train loading is still less established. This thesis investigates the dynamic response of the Åre soil–steel composite culvert in Sweden using field measurements and three-dimensional finite element analysis.

Hammer impact testing and train-passage measurements were performed on both the culvert and the adjacent track section. The processed field data were used to calibrate finite element models in Brigade Plus, including a benchmark track–soil model and a culvert–soil model with a simplified equivalent orthotropic shell representation of the corrugated culvert. The calibrated models were validated against measured responses from an X62 Regina train passage and then used to simulate Eurocode HSLM-A high-speed train loading between 150 and 350 $\frac{\text{km}}{\text{h}}$.

The monitored train passages showed that the ballast response above the culvert and in the adjacent benchmark track section approximately 20 m away were of similar magnitude, while the acceleration measured directly on the steel shell was lower. The validation showed that the models reproduced the general response patterns and dominant dynamic characteristics. However, acceleration and displacement amplitudes were underestimated by approximately a factor of two to three, indicating that the numerical models were stiffer than the real structure.

The high-speed simulations showed that the adjacent track model produced significantly higher acceleration and displacement responses than the culvert model throughout the investigated speed range, likely due to uncertainties in the calibrated backfill soil properties. This was a major limitation of the comparison between the track model and culvert model when assessing acceleration limits. Therefore, the lower acceleration response predicted in the culvert model should not be interpreted as an exact representation of the real system. Instead, it should be treated as an indication that the culvert response is likely not higher than the adjacent track response, while a more accurate conclusion would require improved soil data, embankment geometry, and closer FRF agreement during calibration.

No resonance response was identified from the acceleration–speed curves or the corresponding time-history results. Even when the simulated culvert displacement due to HSLM-A loading was conservatively doubled to account for the validation discrepancy, the maximum response remained below the 2 mm *Trafikverket* serviceability limit. Overall, the results indicate that the Åre culvert is expected to perform satisfactorily under the investigated high-speed train loading, with no clear indication of resonance or excessive displacement.

Keywords: Soil–steel bridge, dynamic analysis, finite element method, field measurements, frequency response function (FRF), high-speed railway traffic, resonance, soil–structure interaction.

Sammanfattning

Jord–stålbroar används ofta för järnvägstillämpningar med korta spännvidder tack vare deras effektiva byggmetod och låga materialåtgång. Deras dynamiska beteende vid höghastighetstågtrafik är dock fortfarande relativt begränsat studerat. I detta examensarbete undersöks den dynamiska responsen hos jord–stålbron i Åre genom fältmätningar och tredimensionell finita elementanalys.

Fältundersökningen omfattade både slaghammartester och mätningar av passerande tåg på såväl bron som den intilliggande spårsträckan. De uppmätta resultaten bearbetades och användes för kalibrering av finita elementmodeller i Brigade Plus. Två modeller utvecklades: en referensmodell för spår–jord-systemet samt en jord–stålmodell där den korrugerade stålkonstruktionen representerades med ett förenklat ortotropt skal. De kalibrerade modellerna validerades mot uppmätta responser från en passage av ett X62 Regina-tåg och användes därefter för simuleringar med Eurocode-lastmodellen HSLM-A för tåghastigheter mellan 150 och 350 $\frac{\text{km}}{\text{h}}$.

Resultaten från de uppmätta tågpassen visade att ballastresponsen ovanför bron och i den intilliggande referenssträckan, cirka 20 meter från brons mittspann, var av liknande storleksordning, medan accelerationen uppmätt direkt på stålkonstruktionen var lägre. Valideringen visade att modellerna kunde återge de övergripande responsmönstren och de dominerande dynamiska egenskaperna. Däremot underskattades accelerations- och deformationsnivåerna generellt med ungefär en faktor två till tre, vilket indikerar att modellerna är något styvare än den verkliga konstruktionen.

Simuleringarna för höghastighetståg visade att referensmodellen för spåret genererade högre accelerationer och deformationer än jord–stålmodellen inom hela det undersökta hastighetsintervallet. Detta bedöms främst bero på osäkerheter i de kalibrerade parametrarna för återfyllnadsmaterialet kring bron. Den lägre respons som erhölls i jord–stålmodellen bör därför inte betraktas som en exakt representation av den verkliga konstruktionen utan snarare som en indikation på att bronsrespons sannolikt inte överstiger responsen i den intilliggande spårsträckan.

Ingen resonans kunde identifieras vare sig i acceleration–hastighetskurvorna eller i de analyserade tidsserierna. Även när de beräknade deformationerna för jord–stålmodellen konservativt fördubblades för att kompensera för skillnaden mellan mätningar och simuleringar, understeg den maximala deformationen Trafikverkets gränsvärde på 2 mm. Sammantaget visar resultaten att jord–stålbron i Åre förväntas uppvisa ett tillfredsställande dynamiskt beteende vid de undersökta höghastighetslasterna, utan tydliga indikationer på resonans eller otillåtna deformationer.

Nyckelord: jord–stålbro, dynamisk analys, finita elementmetoden, fältmätningar, frekvensresponsfunktion, höghastighetståg, resonans.

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Table of Contents

Abstract ii
Sammanfattning iii
Acknowledgments iv
Table of Contents v
List of Figures vii
List of Tables x
List of acronyms and abbreviations xi
1 Introduction 1
1.1 Background 1
1.2 Literature Review 2
1.3 Aim and Scope 4
1.4 Research Methodology 4
1.5 Limitations 5
1.6 Report Outline 5
2 Theoretical Background 6
2.1 Soil-Steel Structure 6
2.2 Equation of motion 7
2.3 Resonance & Rayleigh Velocity 8
2.4 Frequency Domain Analysis 9
2.4.1 Fourier Series Transform 9
2.4.2 Fast Fourier Series Transform (FFT) 9
2.4.3 Signal Sampling & Processing 10
2.5 FRF: Frequency Response Functions 10
3 Case Study Bridge 13
4 Field Measurements & Analysis 16
4.1 Forced Vibration Test: Instrumentation & Loading 17
4.1.1 Train Tracks 17
4.1.2 Steel Culvert 18
4.2 Monitored Train Passages 19
4.3 Modal Analysis 20
4.3.1 Sampling & Signal Processing 21
4.3.2 FRF Analysis 21
5 Finite Element Modeling 22
5.1 Track Model 22
5.2 Orthotropic Culvert Bridge Model 25
5.3 Model Mesh 28
5.3.1 Track model 29
5.3.2 Culvert model 30
5.4 Calibrated Models 32
5.4.1 Track model 32
5.4.2 Culvert model 35
5.5 Model Validation using Passing Trains 40
6 High Speed Train Passage Analysis 46
7 Conclusion and Future work 50
7.1 Conclusions 50
7.2 Future Work 51
References 52

Appendix A: Mesh Convergence 54
Appendix B: C3D to CIN3D FEM Element Conversion 58

List of Figures

Figure 1: Typical layout and soil zones surrounding a corrugated steel culvert, illustrating the structural backfill, bedding layer, road superstructure, and geometrical parameters governing the soil–steel interaction behavior of the culvert system (page 1) [3]. 6
Figure 2: Relationship between time and frequency domain [17]. 9
Figure 4: Field photograph of the Åre culvert showing the corrugated steel pipe arch structure and the surrounding railway embankment. 13
Figure 5: Geometrical parameters defining the cross-sectional shape of the Åre culvert, including the culvert height H, span B, and radii R, R1, and R2. 14
Figure 6: Cross-section of the Åre culvert and surrounding soil layers used in the finite element model. 15
Figure 7: Instrumented modal impact hammer used for the forced-vibration tests conducted on the rail and culvert structure. 16
Figure 8: Accelerometers used for vertical vibration measurements, showing ones at locations A4 (on sleeper) and A3 (in ballast). 17
Figure 9: Instrumentation layout for the experimental testing program, illustrating the placement of accelerometers on the culvert and adjacent track (A1–A13) together with the hammer impact locations (P1–P7). 18
Figure 10: Cross-sectional view of the Åre culvert showing the placement of accelerometers (A1–A2) and hammer impact locations (P0–P2) used during the forced-vibration tests. 19
Figure 11: Elevation view of culvert showing currently operational the X62 Regina train passing by. 20
Figure 12: Three-dimensional finite element model of the benchmark track–soil system developed in Brigade Plus, illustrating the ballast, subgrade soil, and underlying soil layers included in the numerical analysis. 23
Figure 13: Illustration of the added soil and infinite material layers applied at the longitudinal boundaries of the numerical model to reduce artificial wave reflections during the dynamic analysis. 25
Figure 14: Three-dimensional finite element model of the culvert–soil system developed in Brigade Plus, illustrating the railway track, ballast, subgrade soil, underlying soil, backfill soil, underpinning soil, and the equivalent orthotropic steel culvert included in the numerical analysis. 26
Figure 15: Illustration of the equivalent orthotropic shell modelling approach adopted for the corrugated steel culvert, where the corrugated steel profile is represented by an equivalent flat orthotropic shell with equivalent thickness and stiffness properties. 27
Figure 16: Meshed finite element model of the benchmark track–soil system including the added soil and infinite material layers at the longitudinal boundaries. 30
Figure 17: Meshed finite element model of the culvert–soil system including the added soil and infinite material layers at the longitudinal boundaries. 31
Figure 18: Comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated track model for sensors A8 subjected to hammer impact loading at location P7. 33
Figure 19: Comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated track model for sensor A9 subjected to hammer impact loading at location P7. 33
Figure 20: Comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated track model for sensor A11 subjected to hammer impact loading at location P7. 34

Figure 21: Comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated track model for sensor A13 subjected to hammer impact loading at location P7. 34
Figure 22: Comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated culvert model for sensor A1 inside the culvert, subjected to hammer impact loading at location P1-2. 36
Figure 23: Comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated culvert model for sensor A3 subjected to hammer impact loading at location P1-2. 36
Figure 24: Comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated culvert model for sensor A4 subjected to hammer impact loading at location P1-2 37
Figure 25: Comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated culvert model for sensor A1 inside the culvert, subjected to hammer impact loading at location P0-1. 38
Figure 26: Comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated culvert model for sensor A3 subjected to hammer impact loading at location P0-1. 38
Figure 27: Comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated culvert model for sensor A4 subjected to hammer impact loading at location P0-1. 39
Figure 28: First vertical bending mode of the calibrated culvert model obtained from the finite element modal analysis, representing the fundamental vibration mode of the soil–steel culvert bridge. 40
Figure 29: Vertical acceleration time history from recorded data during the passing of X62 Regina Train on Åre Culvert section. 41
Figure 30: Vertical acceleration time history from recorded data during the passing of X62 Regina Train on the track section around 20 m away from Åre Culvert.... 42
Figure 31: Comparison between the experimentally measured and numerically simulated vertical acceleration responses at sensor A12 for the X62 Regina train passage at a speed of 135 km/h. 43
Figure 32: Comparison between the experimentally measured and numerically simulated vertical displacement responses at sensor D1 located at the culvert midspan for the passage of the X62 Regina train at a speed of 135 km/h. 44
Figure 33: Comparison between the experimentally measured and numerically simulated vertical acceleration responses at sensor A1 located at the culvert midspan for the passage of the X62 Regina train at a speed of 135 km/h. 44
Figure 34: Comparison between the experimentally measured and numerically simulated vertical acceleration responses at sensor A3 for the passage of the X62 Regina train at a speed of 135 km/h. 45
Figure 35: Maximum acceleration responses at sensor A1 and A3 (culvert model) and A11 (track model) during the HSLM-A2 train design loads, filtered at 30 Hz. .47
Figure 36: Maximum acceleration responses at sensor A1 and A3 (culvert model) and A11 (track model) during the HSLM-A2 train design loads, filtered at 50 Hz. .47
Figure 37: Maximum displacement responses at sensor A1 and A3 (culvert model) and A11 (track model) during the HSLM-A2 train design loads, filtered at 30 Hz. 48
Figure 38: Acceleration time histories of the track and culvert models filtered at 30 Hz at sensor A1 and A11 for the HSLM-A2 train loading. 49
Figure 39: Mesh sensitivity analysis of the soil domain in the track model showing the frequency response functions (FRFs) obtained for different mesh sizes at a node located on the soil surface beneath the ballast at the track midspan. 55

Figure 40: Mesh sensitivity analysis of the ballast layer in the track model showing the frequency response functions (FRFs) obtained for different mesh sizes at a node located on the ballast surface at the track midspan. 56
Figure 41: Mesh sensitivity analysis of the steel culvert in the culvert model showing the frequency response functions (FRFs) obtained for different mesh sizes at a node located on the top surface of the culvert at midspan. 57

List of Tables

Table 1: Summary of the geometric parameters defining the cross-sectional dimensions of the Åre culvert. 14
Table 2: Summary of the main properties of the train type recorded during field measurements [22]. 20
Table 3: Material properties assigned to the rails, sleepers, and rail pads in the finite element track model. 23
Table 4: Spring directional stiffness connecting the rail-pad elements. 24
Table 5: Equivalent orthotropic steel shell properties. 28
Table 6: Adopted element mesh size and type for the track model. 30
Table 7: Adopted element mesh size and type for the culvert model. 31
Table 8: Calibrated material properties of the track model. 35
Table 9: Calibrated material properties of the culvert model. 39
Table 10: Estimated Rayleigh soil velocities based on calibrated material properties. 39

List of acronyms and abbreviations

$a(f)$	Acceleration response in the frequency domain
$a(t)$	Acceleration response in the time domain
a_o	Mean constant component of the excitation
a_j	Cosine Fourier coefficient of the j-the harmonic
b_j	Sine Fourier coefficient of the j-the harmonic
c	Damping
$C(\omega)$	Frequency-dependent damping matrix
$C_{aF}(f)$	Coherence function between acceleration response and hammer force input
E	Young's modulus
f	Frequency
$F(f)$	Hammer force input in the frequency domain
$F(t)$	Hammer force input in the time domain
f_N	Nyquist frequency
f_s	Sampling frequency
G	Shear modulus
$G_{aa}(f)$	Auto-spectrum of acceleration response
$G_{aF}(f)$	Cross-spectrum between acceleration response and hammer force input
$G_{FF}(f)$	Auto-spectrum of hammer force input
$H_{aF}(f)$	Frequency response function between acceleration response and hammer force input
i	Imaginary unit
j	Harmonic index, positive integer
k	Stiffness
m	Mass of structure
N_s	Number of samples

$P(\omega)$	Force vector in the frequency domain
$p(t)$	Periodic excitation as a function of time
t	Time
T	Period of the excitation
$u(t)$	Displacement response
$\dot{u}(t)$	Velocity
$\ddot{u}(t)$	Acceleration
$U(\omega)$	Displacement vector in the frequency domain
ν	Poisson's ratio
V_R	Rayleigh wave velocity
V_s	Shear wave velocity
Δt	Sampling interval
$(\cdot)^*$	Complex conjugate
Overbar	Average over repeated hammer impacts
ρ	Mass density
ω	Angular frequency
ω_o	Fundamental angular frequency, $\omega_0 = 2\pi/T$

1 Introduction

1.1 Background

Soil-steel composite bridges have become widely adopted in Sweden due to their low environmental and monetary costs in combination with a relatively fast construction schedule for short-span applications. Soil-steel bridge performance, especially in comparison to reinforced-concrete bridges, makes them superior for roadway and railway applications [1]. However, there are several knowledge gaps about their dynamic behavior, particularly under the rapidly growing spread of modern high-speed train traffic. This issue is of growing importance stems from the fact that Sweden is currently planning the construction of high-speed railway lines designed for train speeds of up to 320 km/h, in addition to upgrading existing railway lines to accommodate higher operating speeds [2]. The dynamic response of soil-steel bridges is complex because these bridges combine corrugated steel shells and backfill soil in positive action, in addition to track structures. More specifically, it's the soil–structure interaction that is difficult to model. Estimating the modulus of elasticity of the surrounding soil analytically is challenging as soil stiffness depends on many factors such as compaction state, moisture level, stress history, and strain amplitude. Simple analytical models often cannot capture the actual in-situ behavior [1].

The design approach for soil steel structures used in Sweden is the approach proposed by Pettersson and Sundquist [3], in addition to provisions for dynamic loading and railway design specified by Eurocode and *Trafikverket*. Eurocode guidelines state that the vertical bridge deck acceleration is limited to 3.5 $\frac{m}{s^2}$, and the vertical bridge deflection caused by the live loads should not exceed 8 mm [4][5]. However, per the Swedish guidelines, *Trafikverket*, the maximum allowable vertical displacement at the level of the underside of the sleepers in train tracks shall be 2 mm, peak-to-peak, during train passage at the permitted line speed [6]. It's important to note that both the EU code and Swedish guidelines for dynamic analysis and railways specify load models and performance limits; however, they do not provide guidance on soil stiffness or damping for soil–steel composite bridges specifically, which further supports the need for field-calibrated numerical models for this project.

Thus, field testing of dynamic behavior for such structures offers a more reliable method for parameter estimation compared to analytical or laboratory-scale methods. Measuring the response directly in the field, from controlled impact testing, can allow the back-analysis of dynamic properties such as eigenfrequencies, damping ratios and effective stiffness. These measured parameters can then be used to calibrate the finite element model. High speed trains cause high loads, rapid load cycles, possible resonance effects, and soil–structure interaction that may affect fatigue, serviceability, and safety. While there are many previous studies focused on static or low-speed loading conditions, the dynamic response of soil-steel bridges to high-speed train load remains less explored. The purpose of this master thesis is to combine field measurement and finite element modelling to provide a

relatively accurate prediction of the dynamic behavior of a soil–steel composite bridge structure, located in Åre, Sweden, under high-speed train loading.

1.2 Literature Review

Beben (2020) [1] explains that soil–steel bridges are interaction-dominated systems, and that simplified analytical approaches often fail to reproduce measured behavior because of soil-steel interaction (SSI) complexity and importance of accurately capturing backfill quality in analysis. The analysis of dynamic parameters in detail, such as backfill impact on damping and dynamic response, can contribute to the optimization of soil cover needed and to an increase in acceptable spans under traffic loads. The discrepancies found in previous research between calculation and experiment motivate advanced numerical and experimental research. Field investigations of soil–steel composite bridges reported by Beben emphasize that reliable dynamic assessment requires simultaneous measurement of displacements and accelerations at both the steel shell and in the overlying ballast and track. Field testing should prioritize vertical displacement at the crown and vertical acceleration in the ballast and track zone. These measured parameters capture both the global soil–structure interaction response in addition to the local dynamic response of the backfill soil around the steel culvert.

In terms of numerical modelling, Beben's research also recommends that finite element analyses should be validated primarily against measured peak displacements, peak accelerations in the adjacent track region, and dominant frequency content. Beben's results also further indicate that effective damping under live train loading is significantly higher than that obtained from ambient vibration tests, highlighting the importance of considering load-dependency in measured values. Several other investigations regarding dynamic analysis of soil steel structures, such as Andersson (2020) [7], Mellat (2014) [8], and Flener (2009) [10], presented results that are overall in agreement with the findings from Beben.

Andersson (2020) investigated the dynamic feasibility of short-to-medium span soil-steel composite bridges (SSCB) for high-speed railway applications using a combined experimental and numerical approach [7]. Findings highlighted that both natural frequency and damping are amplitude-dependent, likely due to the nonlinear behavior of the soil-structure system. The research utilized a model updating algorithm to calibrate 3D finite element models against field data and developed a computationally efficient 1D track model for high-speed simulations. The results demonstrated that these structures, ranging in span from 3.7-4.9 m, are likely to meet the vertical deck acceleration requirements stipulated by Eurocode [4] for train speeds up to 400 km/h. In addition, the tests showed that the double track bridge structures, in comparison to the single-track structures, exhibited a higher system stiffness. The most important takeaways from this study were that the dynamic response of soil–steel composite bridges is governed by soil–structure interaction, that measured damping is significantly higher than typically assumed in analytical models, and that uncalibrated numerical models tend to overestimate resonance effects. Additionally, this investigation compared the results between a corrugated steel culvert model and a simplified orthotropic deck

model. Overall, the simplified model was found to provide sufficiently accurate predictions of the dynamic response of soil–steel composite bridges, while significantly reducing computational cost.

Mellat et al. (2014) and Andersson (2012) investigated the dynamic behavior of the same short-span soil–steel composite railway bridge, with a span of 3.75 m and a backfill height of 1.7 m, using full-scale field measurements and finite element modelling. The bridge was evaluated under regular train passages and high-speed railway loading, with simulations performed for train speeds up to approximately 320 $\frac{km}{h}$. Field measurements showed that the vertical displacements at the culvert crown were small, generally remaining below 0.5 mm, while the acceleration response was more significant. Vertical accelerations at the culvert crown were reported to range between approximately 0.8 and 1.4 $\frac{m}{s^2}$, while higher accelerations were observed at the ballast and sleeper level, reaching up to about 3.5 $\frac{m}{s^2}$. This indicates that acceleration response, particularly at track level, is more critical than displacement when assessing the serviceability performance of soil–steel composite bridges under railway loading. The numerical analyses showed that both simplified orthotropic shell modelling and full three-dimensional finite element modelling could reproduce the general dynamic behavior of the structure, although the 3D model provided better agreement with measured track accelerations. The studies also indicated that no clear resonance behavior was observed under the investigated train passages, although localized dynamic amplification occurred at certain speeds. Overall, the findings suggest that soil–steel composite bridges can remain within serviceability limits under high-speed railway loading, while also highlighting the importance of accurately representing soil stiffness, damping, track-level response, and soil–structure interaction in numerical models [8] [9].

In contrast to studies focusing on short-span bridges under very high train speeds, Flener (2009) [10] evaluated the dynamic response of a long-span SSCB with an 11 m span and a 1.78 m backfill under locomotive speeds ranging from 10 to 125 $\frac{km}{h}$. Field measurements yielded a maximum vertical crown displacement of 0.86 mm at the highest tested speed. The study indicated that the dynamic response and displacement exceeded static responses by up to 20%, while the Dynamic Amplification Factor (DAF) for bending moments at the quarter point reached 1.45, significantly higher than the values recorded at the crown. In a follow-up study of the same bridge, Flener (2010) [11] compared seasonal test results and found that crown displacements were larger in May than in October, suggesting the structure became more flexible after the first winter. This study demonstrated that locomotive speed and braking significantly impacted dynamic response, with full-power braking increasing the amplification factor by 5% [11]. In compliance with established safety criteria, measured ballast accelerations were found to be noticeably lower than the Eurocode limits of 3.5 $\frac{m}{s^2}$, indicating a low risk of ballast instability. Similar to the findings of Mellat et al. (2014) and Andersson (2020), the damping ratios were shown to be strongly amplitude-dependent, increasing from 2% during ambient vibration to 20% during forced vibration.

Other dynamic analysis investigations performed for normal passenger and freight train loads, such as Beben (2018) [12], showed a maximum measured vertical deflection at the crown below 0.65 mm, while peak vertical accelerations reached 1.31 $\frac{m}{s^2}$ in the ballast and 0.71 $\frac{m}{s^2}$ at the bridge crown,

remaining well below the Eurocode limit. Frequency-domain analysis of displacement signals identified dominant response frequencies in the range of 0.3–1.25 Hz, which were largely independent of train type. The results also showed that dynamic response levels were governed more by train weight than speed, and that soil cover provided significant reduction of vibrations in comparison to vibrations at the ballast level.

1.3 Aim and Scope

The following is a summarized list of the aim and scope of this thesis:

- Investigate the dynamic response and properties of the soil-steel bridge and adjacent rail tracks, using full-scale experimental testing, under forced vibrations from impact loading.
- Investigate the dynamic response and properties of the soil-steel bridge and adjacent rail tracks under currently running train loads.
- Create a 3D FEM in Brigade Plus of the existing culvert and adjacent rail tracks section and calibrate the soil material properties in the model based on in-situ field derived data.
- Investigate the difference in structural stiffness and overall dynamic response between the adjacent track section and the soil-steel culvert section.
- Simulate high speed (HSLM-A) train loads on the calibrated FEM to predict the dynamic behavior of the structure under high-speed train loading and evaluate the risk of resonance under modeled loads.

This research aims to investigate the dynamic response of the soil-steel composite bridge when subjected to high-speed railway traffic. Focusing on expected acceleration and displacement amplitudes, soil-steel interaction, and potential speed-dependence. Overall, the objective is to improve understanding of dynamic behavior and suitability of soil-steel composite bridges under high-speed railway loading, especially considering the low profile backfill and relatively large span of this bridge case study. Particular attention is given to modelling assumptions related to soil stiffness, damping, and boundary conditions when performing finite element modeling.

1.4 Research Methodology

The scope of this master thesis creates a methodological design that includes field experiments, modal analysis, and finite element modeling and simulation. For the field experiment phase a full-scale soil–steel composite bridge, Åre Culvert, and adjacent rail track were instrumented with accelerometers and displacement transducer (LVDT). Forced harmonic vibrations were then introduced by using a hammer impact loading that was applied at defined locations. Measurements from the culvert and the adjacent train track were recorded for the forced impact loading, and additionally for currently passing trains.

Following this, a frequency response function modal analysis was performed using MATLAB to determine the system's response to various excitation frequencies from impact loading and passing

trains. The finite element modelling phase included a simplified orthotropic model of the culvert together with the adjacent rail track section. Results from the field testing and analysis were then used to calibrate the numerical models by adjusting the soil material properties in a three-dimensional finite element model developed in Brigade Plus. The calibration process was carried out in a stepwise manner, where the adjacent rail track model was calibrated first, followed by the culvert model. To verify the validity of the calibrated models, a validation step was performed using responses obtained from passing trains. The calibrated culvert model was then subjected to high-speed train loading, applied as a moving point load based on the HSLM-A model specified in Eurocode [4], representing a high-speed train passage. The resulting peak acceleration and displacement responses were evaluated to assess the dynamic behavior of the system.

1.5 Limitations

The following limitations apply to this thesis:

- This study did not include seasonal experimental testing that considers temperature changes.
- Only linear soil behavior was considered in the finite element analyses. While soils may exhibit nonlinear behavior under dynamic loading, the small displacement amplitudes observed during train passages indicate that a linear response can reasonably be assumed.
- An equivalent orthotropic shell was modeled instead of corrugated shell in the finite element model.
- The finite element model was calibrated based on assumed full interaction between the steel shell and adjacent soil.
- No information about the geometry of the soil embankment in the track section was unavailable. Therefore, the soil layers in the adjacent track system were estimated during model development and calibration.

1.6 Report Outline

Chapter 2 presents relevant theoretical background for key areas in this research including concepts of soil-steel structure design and behavior, Structural and Soil dynamics, Receptance Modal Analysis, and finite element method application. Chapter 3 presents the soil-steel culvert case study, Åre Culvert, including the culvert and track geometry and material properties. Chapter 4 describes the field measurements and analysis performed, concerning instrumentation, loading, and data processing and filtering. Chapter 5 describes the 3D numerical models created using Brigade Plus. In addition, Chapter 5 presents the process of model calibration and validation based on field derived data and explains modeling choices made. Chapter 6 presents the findings from high-speed train load simulations applied to the calibrated culvert and track models. Finally, chapters 7 and 8 present discussion section, conclusions, and final reflections.

2 Theoretical Background

2.1 Soil-Steel Structure

The structure capacity of soil-steel systems, such as corrugated steel culvert bridges, is mainly governed by the interaction between the surrounding soil and the corrugated steel shell, where the steel shell and the compacted backfill act together as a composite system. The soil provides confinement and support, thereby significantly enhancing the load-carrying capacity overall stability of the structure [7].

The interaction is presented through interfacial forces, consisting of normal and tangential components acting along the surface of the steel shell. These forces play a critical role in ensuring structural integrity and effective distribution of load within the composite system [14]. Furthermore, the mechanical properties of the material involved, particularly the elastic modulus of the soil, have a significant influence on the load-bearing capacity of the system. The soil must therefore possess sufficient strength and stiffness to safety support the applied traffic loads. In addition, the soil should exhibit properties to minimize the risk of frost-induced damage to both the overlaying railway and the structure itself [3].

To better understand the behavior of soil–steel composite culverts, it is important to recognize that the surrounding soil is an essential structural component of the system. Unlike conventional rigid bridges, the load-carrying capacity of corrugated steel culverts is highly dependent on the interaction between the steel shell and the surrounding compacted soil. The backfill, bedding, and surrounding soil layers contribute significantly to the distribution of loads, confinement of the steel shell, and the overall stiffness and stability of the structure, see Figure 1. Consequently, the quality, geometry, and mechanical properties of the surrounding soil have a direct influence on the soil–steel interaction behavior and the structural response of the culvert system [3].

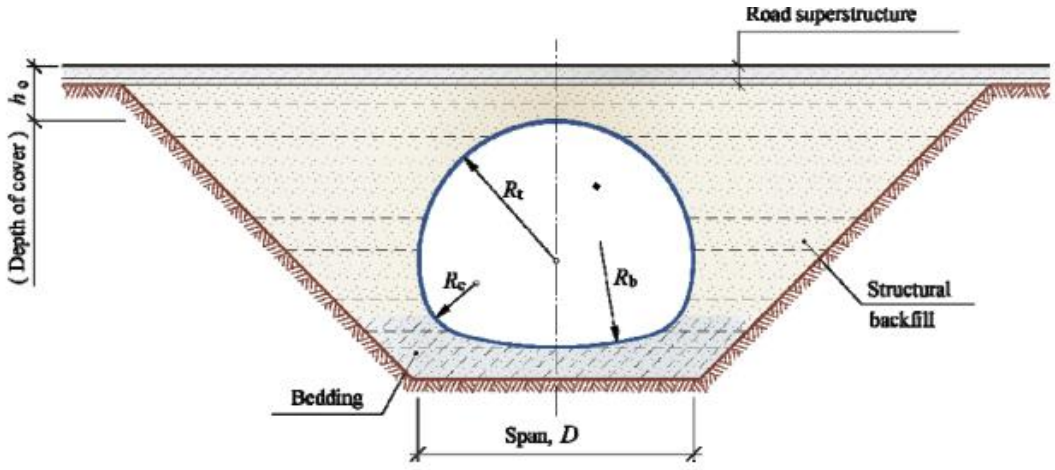


Figure 1: Typical layout and soil zones surrounding a corrugated steel culvert, illustrating the structural backfill, bedding layer, road superstructure, and geometrical parameters governing the soil–steel interaction behavior of the culvert system (page 1) [3].

2.2 Equation of motion

The dynamic response of a structural system subjected to time-dependent loading can be described using the equation of motion, which expresses the equilibrium between inertia, damping, elastic restoring forces, and external dynamic excitation. For a single-degree-of-freedom (SDOF) system, where the motion is restricted to specific direction or plane, and all other possible movements are negligible, this equation is given as:

$$m\ddot{u}(t) + c\dot{u}(t) + ku(t) = p(t) \tag{1}$$

Where m is the lumped mass of the system, c is the damping coefficient, and k is the stiffness. The term $u(t)$ represents the displacement response, while $\dot{u}(t)$ and $\ddot{u}(t)$ denote the velocity and acceleration, respectively. The function $p(t)$ represents the externally applied dynamic load [15].

This equation represents the dynamic equilibrium of the system, where the internal force, the damping force, and the elastic restoring force together balance the applied excitation. The inertial term accounts for resistance to acceleration, the damping term models energy dissipation mechanisms such as material damping and friction, and the stiffness term represents the elastic resistance of the structure against deformation. By solving this equation, key response quantities sch as displacement, velocity, acceleration and internal forces can be obtained.

However, for complex structural systems such as steel-soil bridges, multiple directions of motion are required to accurately represent the dynamic response of the structure when subjected to time-dependent loading. For structure with multiple degrees of freedom (MDOF), the dynamic response is governed by the matrix equation of motion as:

$$[-\omega^2 M + i\omega C(\omega) + K]U(\omega) = P(\omega) \tag{2}$$

Where **ω** donates the natural circular frequency of the structure, M is the mass matrix, i is the imaginary unit, $C(\omega)$ represents the frequency-dependent damping matrix, K is the stiffness matrix, and $U(\omega)$ and $P(\omega)$ denote the displacement and force vectors in the frequency domain, respectively [15].

The dynamic behavior of a structure system is fundamentally governed by its mass, stiffness and damping properties. These parameters collectively determine how a structure responds to time-dependent loading, including its natural frequencies and vibration response. Mass represents the inertial resistance of the structure to acceleration, while stiffness defines its elastic resistance to deformation. These two properties together control the natural vibration characteristics of the system. For single-degree-of-freedom system, the natural circular frequency is given by:

$$\omega_n = \sqrt{\frac{k}{m}} \tag{3}$$

This relationship shows that the natural frequency increases with increasing stiffness and decreases with increasing mass. A stiffer structure vibrates at higher frequencies, while a heavier structure vibrated more slowly.

2.3 Resonance & Rayleigh Velocity

Resonance in structural dynamics refers to the instance of amplified vibrations that occur when the natural frequency of the structure coincides with the frequency of an external force. When analyzing bridge structures, resonance is crucial to consider if the bridge is expected to carry high speed train loads, due to the potential for large vibrations that can compromise structural integrity and safety [15]. Resonance due to axle load repetition from train loads can be expressed in terms of a critical speed at which the load repetition frequency matches a structural eigenfrequency.

In soil-steel bridge structures, assessment of resonance concerns the response of the top soil layers surrounding the steel culvert. In soil dynamics and railway vibration studies, a critical speed phenomenon is observed when the velocity of a moving load approaches the Rayleigh wave velocity of the supporting soil, leading to wave accumulation and amplified ground response. Rayleigh wave velocity is the propagation speed of surface waves in soil and is governed by the shear modulus, mass density, and Poisson's ratio [15] It can describe how train-induced vibrations propagate through the soil. The following equations are used to calculate the speed to check whether ground-wave resonance or amplification can occur:

Shear wave velocity is:

$$V_s = \sqrt{\frac{G}{\rho}} \tag{4}$$

Shear wave velocity expressed in terms of E:

$$V_s = \sqrt{\frac{E}{2\rho(1+\nu)}} \tag{5}$$

Rayleigh wave velocity (approximation):

$$V_R \approx V_s \frac{0.862 + 1.14\nu}{1+\nu} \tag{6}$$

2.4 Frequency Domain Analysis

2.4.1 Fourier Series Transform

A Fourier series represents a periodic function as a sum of harmonic sinusoidal components with different frequencies. Any periodic excitation can be expressed as a series of sine and cosine terms whose frequencies are integer multiples of the fundamental frequency [15].

$$p(t) = a_0 + \sum_{j=1}^{\infty} a_j cos(j\ \omega_0 t) + \sum_{j=1}^{\infty} b_j sin(j\ \omega_0 t) \tag{7}$$

In dynamic analysis, these frequencies are commonly expressed in terms of angular frequency as follows:

$$\omega = 2\pi f \tag{8}$$

2.4.2 Fast Fourier Series Transform (FFT)

The Fourier transform provides a mathematical tool that can be used for converting a signal from the time domain into the frequency domain, as shown in Figure 2. The frequency-domain representation reveals information that is not apparent in the time domain, such as dominant frequencies, harmonics, and resonance phenomena.

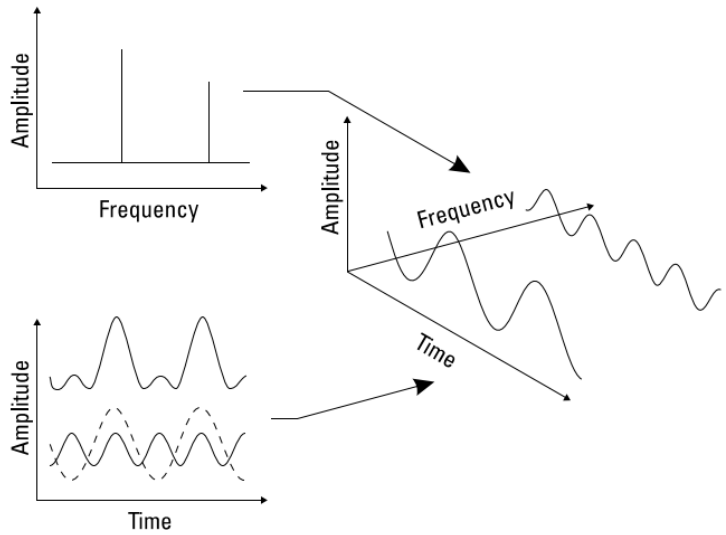


Figure 2: Relationship between time and frequency domain [17].

Signal transformation can be performed through The Discrete Fourier Transform (DFT), which is the numerical application of the Fourier transformation. It is used to analyze the frequency content of discretely sampled signals. The sampled time signal in this case is decomposed into a finite set of

discrete frequency components, each representing the amplitude and phase of a sinusoidal component present in the signal as follows:

$$F(\omega) = \frac{1}{N}\sum_{0}^{N-1} f(k)e^{-j\omega kT} \quad (9)$$

where ω is the natural frequency, N is the number of samples, k are the discrete samples of the original signal, j is the imaginary number and T is the period.

The Fast Fourier Transform (FFT) is an efficient numerical algorithm used to compute the discrete Fourier transform of sampled time-domain data. Rather than performing the Fourier transform directly, which would require many computations, the FFT significantly reduces the computational effort by using signal symmetry feature in the transformation process. The FFT function operates on a specific time range consisting of equally spaced samples and transforms this data block into a set of discrete frequency lines representing amplitudes and phases [17].

2.4.3 Signal Sampling & Processing

Since the FFT processes the entire time record simultaneously, the frequency resolution depends on the length of the time record, and the maximum resolvable frequency depends on the sampling rate. The number of samples used in the analysis strongly influences the frequency resolution of the Fourier transform. The frequency resolution is inversely proportional to the total duration of the sampled signal, meaning that longer time records provide smaller separation between adjacent frequency lines. One of the problems that can happen is spectral leakage. The Fourier transform assumes that the sampled time record is periodic, which is not always the case for measured signals. This could lead to a sudden truncation of the signal at either the start or end of the sampling window. The windowing function tapers the signal smoothly to zero at the boundaries, thereby reducing leakage and improving the accuracy of the reconstructed frequency spectrum [17].

Additionally, the choice of sampling frequency is fundamental in digital signal processing because it determines the maximum frequency content that can be accurately represented in a sampled signal. According to the Nyquist–Shannon sampling theorem, the sampling frequency must be at least twice the highest frequency present in the signal to avoid the problem of aliasing as follows:

$$f_{max} = f_{Nyquist} = \Delta f\left(\frac{N}{2}\right) = \frac{1}{2\Delta t} \quad (10)$$

Filtering is an essential preprocessing step in signal analysis to remove frequency components that are not relevant to the analysis or that may corrupt the results. Anti-aliasing filters are applied prior to sampling to eliminate high-frequency content above the Nyquist frequency, thereby preventing aliasing effects.

2.5 FRF: Frequency Response Functions

In structural dynamics, Frequency response Functions (FRFs) describe the relationship between an applied dynamic excitation (input) and the resulting structural response (output) in the frequency

domain. Instead of expressing response as a function of time, the governing equation of motion is transformed using Fourier analysis, allowing the response to be expressed as a function of excitation frequency. This formulation is particularly useful for analyzing steady-state vibration, resonance behavior, and dynamic amplification under harmonic loading.

According to Chopra [15], the frequency-domain form of the equation of motion for a SDOF system, as defined in Eq (1), subjected to harmonic excitation leads to a direct relationship between the transformed displacement response $U(\omega)$ and the transformed force $F(\omega)$. This relationship is expressed through a frequency response function, defined as:

$$H_u(\omega) = \frac{U(\omega)}{F(\omega)} = \frac{1}{-m\omega^2 + ci\omega + k} \tag{11}$$

Where m is the mass, c is the viscous damping coefficient, k is the stiffness of the system, ω is the circular excitation frequency, i and is the imaginary unit. The denominator of this expression represents the system's dynamic stiffness, accounting for inertia, damping and elastic restoring forces.

In the frequency domain, velocity is related to displacement through the first time derivative, which gives:

$$\dot{U}(\omega) = -\omega U(\omega) \tag{12}$$

Similarly, acceleration is related to displacement through the second time derivate, which yields:

$$\ddot{U}(\omega) = -\omega^2 U(\omega) \tag{13}$$

Using this relationship, the acceleration response per unit applied force, can be derived as follows:

$$A(\omega) = \frac{\ddot{U}(\omega)}{F(\omega)} \tag{14}$$

While the above formulation represents an analytical frequency-domain derivation, the experimental part of this study is based on field-measured-time-domain data. The recorded signals are transformed into the frequency domain using the Fast Fourier Transform (FFT), allowing frequency response functions to be obtained from experimental measurements.

For this study, repeated hammer impact was applied per sensor location. Thus, the acceleration-based frequency response function was calculated as the ratio between the averaged cross-spectrum of the acceleration response and hammer force input, and the averaged auto-spectrum of the hammer force. This gives the acceleration response per unit applied force as:

$$H_{aF}(f) = \frac{G_{aF}(f)}{G_{FF}(f)} \tag{15}$$

Where,

$$G_{aF}(f) = a(f)F^*(f) \tag{16}$$

$$G_{FF}(f) = F(f)F^*(f) \tag{17}$$

Where:

- $H_{aF}(f)$= acceleration response per unit hammer force
- $a(f)$= acceleration response in the frequency domain
- $F(f)$= hammer force input in the frequency domain
- $F^*(f)$= complex conjugate of the force spectrum
- $\bar{(.)}$= averaging over the repeated hammer impacts

The coherence function was calculated, as shown in Eq (18), between the acceleration response a and the hammer force input F. The coherence value varies between 0 and 1. A coherence close to 1 indicates that the acceleration response at that frequency is strongly related to the hammer force input. A low coherence value indicates that the response may be affected by noise, nonlinear behavior, poor impact quality, or other unmeasured excitation sources.

$$C_{aF}(f) = \frac{\left|\overline{G_{aF}(f)}\right|^2}{\left[\overline{G_{aa}(f)}\right]\left[\overline{G_{FF}(f)}\right]} \tag{18}$$

Where:

$$G_{aa}(f) = a(f)a^*(f) \tag{19}$$

3 Case Study Bridge

The bridge investigated in this thesis is the Åre Culvert, a closed-shaped corrugated steel culvert located in Åre, Sweden, which was constructed in 2019, see Figure 4. The site is situated approximately 525 km northwest of Stockholm and forms part of the Swedish railway network operated by *Trafikverket*. Åre Culvert is a single-track soil–steel composite bridge, classified as Type 1620 according to the BatMan codebook.



Figure 3: Field photograph of the Åre culvert showing the corrugated steel pipe arch structure and the surrounding railway embankment.

The structure consists of a corrugated steel pipe arch that interacts with the surrounding soil to resist external loads through soil–steel interaction. The culvert has an elliptical cross-section with a vertical diameter of 6.54 m, a horizontal diameter of 7.52 m, and a span of 7.46 m. The steel shell is covered by a relatively shallow backfill, with a crown height of 0.5 m between the top of the culvert and the underside of the ballast.

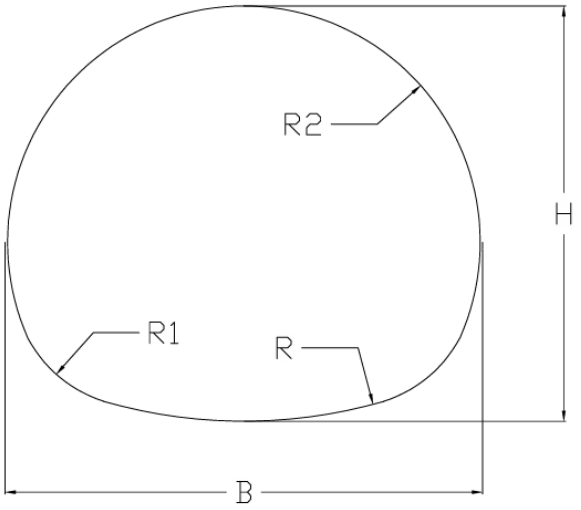


Figure 4: Geometrical parameters defining the cross-sectional shape of the Åre culvert, including the culvert height H, span B, and radii R, R1, and R2.

The culvert is constructed using corrugated structural steel profiles of 200 × 55 mm, with a corrugation depth of 0.055 m and a pitch of 0.20 m. The steel shell has a cross-sectional area of 22.56 m² and is manufactured from structural steel grade S355. The thickness of the steel plates varies between 5 mm and 6 mm, where the crown section consists of 6 mm plates and the remaining sections consist of 5 mm plates. The geometry of the culvert, including the defining cross-sectional parameters, is illustrated in Figure 5, while the corresponding dimensions are summarized in Table 1.

Table 1: Summary of the geometric parameters defining the cross-sectional dimensions of the Åre culvert.

Parameter	**R**	**R1**	**R2**	**H**	**B**
Value [m]	7,948	1,885	3,273	6,54	7,40

The surrounding soil system is divided into four distinct zones, including the backfill and foundation layers around the steel structure, as illustrated in Figure 6.

The bridge is designed for railway traffic loads according to Load Model LM71. The railway line consists of a single track, and the maximum permitted train speed at the site is approximately 120 km/h. In accordance with Swedish bridge design regulations, the structure is designed for a technical service life of 120 years.

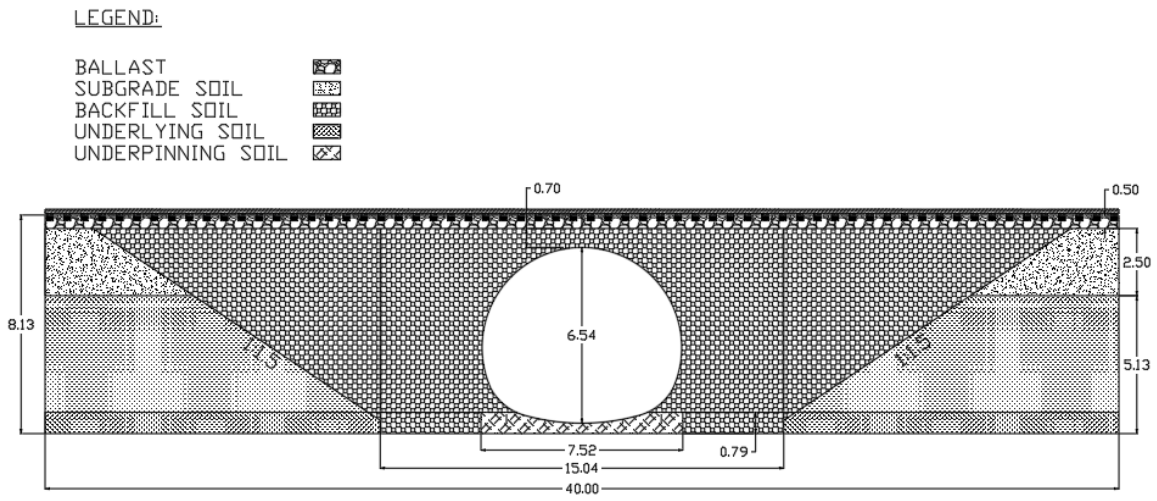


Figure 5: Cross-section of the Åre culvert and surrounding soil layers used in the finite element model.

4 Field Measurements & Analysis

To study and understand the dynamic response of the steel-soil interaction of the Åre culvert, an experimental testing program was conducted on site in coordination with *Trafikverket*. The objective of these measurements was to capture the vibration characteristics of both the culvert structure and the surrounding track system under controlled and operational loading conditions.

The field measurements consisted of the following tests for both the culvert and the adjacent track system:

- Forced-vibration tests using hammer impact loading.
- Train-passage measurements during regular train operations.

The forced vibration tests were conducted using an instrumented modal impact hammer equipped with an integrated force transducer, as shown in Figure 7. The hammer was used to apply controlled impulse loads at predefined impact locations on the culvert and track system. The force transducer in the hammer head measured the applied impact force as a time-history signal, while accelerometers installed at selected locations recorded the corresponding vertical acceleration response of the structure, as shown in Figure 8. Additionally, a displacement transducer (LVDT) was placed on the inside of the steel culvert to record vertical displacement of the structure.

The measurement data were recorded using the CATMAN AP data-acquisition software [18]. The software was used to monitor the signals during testing and store the force and acceleration time histories for subsequent post processing. For each impact location, several repeated hammer impacts were applied to improve the reliability of the frequency response estimates. The recorded hammer force signal was later used as the input excitation, while the measured acceleration signals were used as the output responses in the frequency response function analysis.



Figure 6: Instrumented modal impact hammer used for the forced-vibration tests conducted on the rail and culvert structure.



Figure 7: Accelerometers used for vertical vibration measurements, showing ones at locations A4 (on sleeper) and A3 (in ballast).

4.1 Forced Vibration Test: Instrumentation & Loading

For the impact hammer test the procedure was repeated ten times at each selected impact point. The response spectrum for each location was then averaged during post processing of results. This process was done to help reduce variation and random noise that could happen between the individual hammer hits.

4.1.1 Train Tracks

The testing on the structure began with the installation of accelerometers at predefined locations based on the culvert geometry. Sensors were installed on the railway track above the culvert and along the track in the direction SL spår Duved, as shown in Figure 9.

- Sensor A3 was placed in the ballast at the culvert midspan.
- Sensor A4 was installed on the sleeper adjacent to midspan, in the direction of *SL spår Duved.*
- Sensors A5 and A7 were positioned in the ballast

To capture the dynamic response further away from the culvert, additional sensors were installed along the track around 20 m away from midspan:

- Sensor A8 was placed at approximately 17.88 m from the culvert midspan, corresponding to sleeper number 28.

- Sensors A9, A11, A12, and A13 were installed in the ballast, extending further in the direction of *SL spår Duved*.

Following, impact hammer loading was applied at predefined locations along the track, as shown in Figure 9:

- P4 was applied on the rail, located at sleeper number 30 from midspan culvert.
- P5 was applied on the sleeper number 30 from midspan culvert.
- P6 was applied on the rail, located at sleeper number 32 from midspan culvert.
- P7 was applied on the rail, located at sleeper number 34 from midspan culvert.

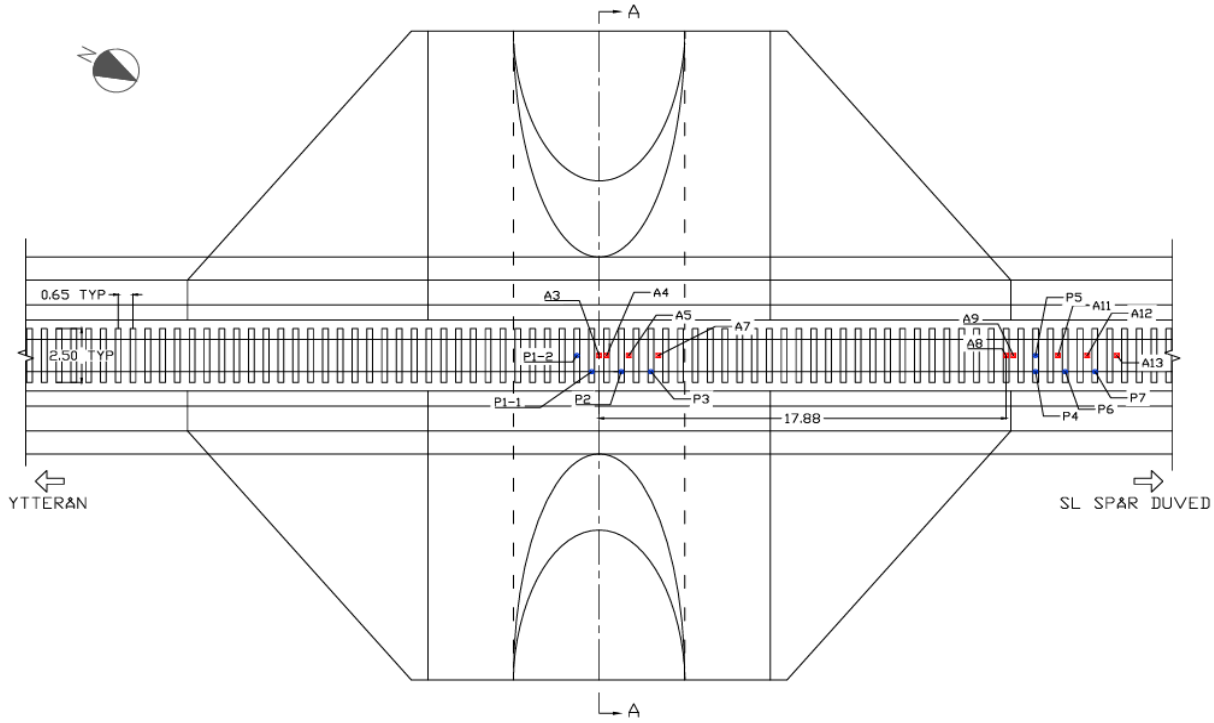


Figure 8: Instrumentation layout for the experimental testing program, illustrating the placement of accelerometers on the culvert and adjacent track (A1–A13) together with the hammer impact locations (P1–P7).

4.1.2 Steel Culvert

In addition, similar impact testing was perforce on the steel culvert. Sensor positions were determined by measuring the culvert span and identifying key points along its length in relation to first mode bending, see Figure 10:

- Sensor A1 was installed at the midspan of the culvert.

- Sensor A2 was positioned at 3.5 m towards the east direction from midspan, as illustrated in.

Following instrumentation, the locations for the impact hammer loading were defined:

- Impact point P0-1 was placed adjacent to sensor A1, at a distance of two full corrugation waves (approximately 40 cm) from the accelerometer.
- Impact point P0-2 was placed adjacent to sensor A1, at a distance of two full corrugation waves.

All impacts were applied on the lower corrugation of the culvert, ensuring consistent excitation of the dynamic response.

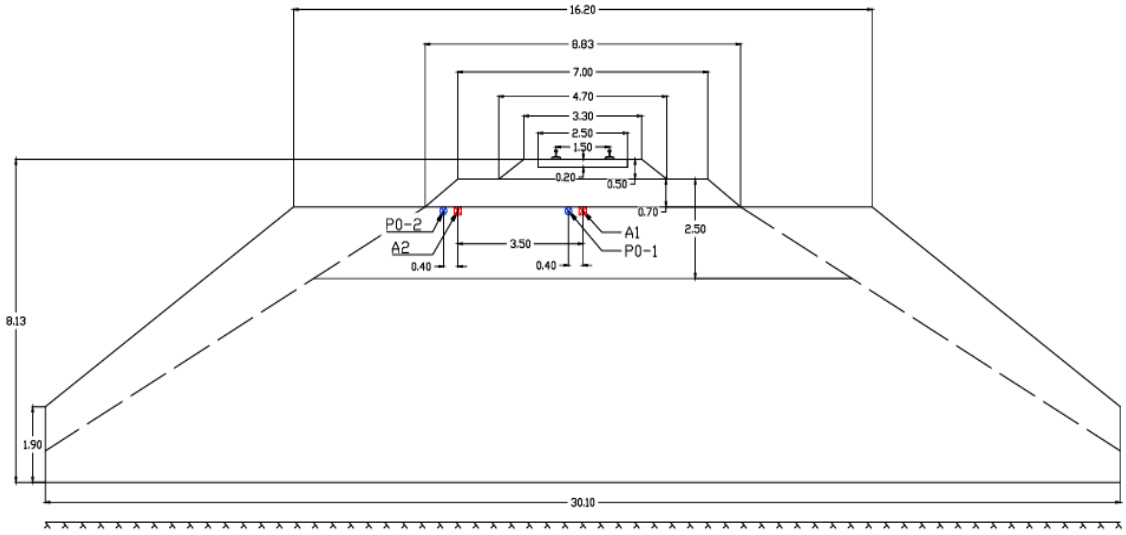


Figure 9: Cross-sectional view of the Åre culvert showing the placement of accelerometers (A1–A2) and hammer impact locations (P0–P2) used during the forced-vibration tests.

4.2 Monitored Train Passages

In addition to controlled excitation, the dynamic response of the culvert structure was monitored and recorded during regular train passages, as shown in Figure 11 for X62 Regina train. The characteristics of the monitored train type are summarized in Table 2. The listed parameters describe the global train configuration and loading conditions, including axle arrangement and maximum axle load. The axle arrangement for the X62 Coradia Nordic is four-car articulated EMU with powered and one unpowered bogie group. The measurements were carried out under the supervision of *Trafikverket* to ensure safe operation and accurate timing of the recordings. These train-passage measurements provide information on the dynamic response of the culvert and surrounding soil under real operating conditions and were later used to support the validation of the numerical model.

the response of the culvert and railway track during train passages was measured using sensors (A1–A13), as illustrated in Figures 9 and 10. The vertical displacement during passing train loads was measured using LVDT that was placed on the steel culvert at midspan, at two corrugations away from the location of accelerometer A1. The recorded vertical acceleration and displacement from train passage was later used to validate the calibrated models by simulating the same train loads in the model and comparing the calibrated numerical model response to the recorded actual bridge response.



Figure 10: Elevation view of culvert showing currently operational the X62 Regina train passing by.

Table 2: Summary of the main properties of the train type recorded during field measurements **Error! Reference source not found.**.

Train type	Configuration	Length (m)	Axle arrangement	Service weight (tons)	Max axle load (kN)	Max speed ($\frac{km}{h}$)
X62 Coradia Nordic	Four-car EMU	74.3	Bo'(Bo')(2')(Bo')Bo'	155	201	180

4.3 Modal Analysis

The purpose of modal analysis was to identify the dominant resonance frequencies of the track and culvert structures and to obtain FRFs that are later used to calibrate the finite element model. The measured input is the force from hammer impact, and the output is the recorded vertical acceleration time history at the specified sensor location. The MATLAB scripts used for post-processing the recorded hammer-impact and train-passage data were provided from an ongoing KTH PhD research project on soil–steel composite bridges. The scripts were not developed as part of this master thesis,

but were used to process the measured time histories, calculate FRFs, and evaluate coherence for comparison with the finite element models.

4.3.1 Sampling & Signal Processing

The measured hammer-test data was stored as MATLAB files. Each file contained the time vector, the sampling frequency, the impact force signal, and the acceleration response signals for the consecutive 10 impacts for each specified accelerometer. The sampling frequency of the measurements was 1200 Hz, corresponding to a time increment of 0.000833 s. This gives a Nyquist frequency of 600 Hz, meaning that frequencies up to 600 Hz can theoretically be represented without aliasing. Since the frequency range of interest in this study was limited to 1–30 Hz, the selected sampling frequency was sufficiently high to capture the relevant dynamic response of the system.

The time-domain acceleration responses were first plotted for all sensors. Following, individual hammer impacts were identified using find peaks function in MATLAB. After identifying the impacts, the script plotted the force pulses and their maximum amplitudes to verify the repeatability of the hammer hits.

4.3.2 FRF Analysis

The frequency response function gives the acceleration response per unit input force. In this study, detected force peak was treated as one hammer impact and the corresponding time segment around each impact was extracted and used in the frequency-domain analysis.

Following, for each impact segment the force signal and acceleration response are transformed from time domain to the frequency domain using Fast Fourier Transform, FFT, function in MATLAB. For every sensor, the frequency spectrum is calculated for each hammer impact and then averaged over ten repeated impacts.

The FRF was calculated from the ratio between the average spectrum of the acceleration response and hammer force, and the average spectrum of the hammer force. This input-normalized FRF estimate accounts for variations in impact force amplitude between repeated hammer hits. Following the coherence function was calculated to evaluate the quality of the FRF estimate where the magnitude of the FRF and the coherence were plotted together for each sensor location.

5 Finite Element Modeling

5.1 Track Model

To capture the dynamic soil-steel interaction (SSI) behavior of the bridge system, a 3D-numerical model representing only the track-soil configuration was first developed in Brigade Plus. The track-only model serves as a benchmark model, establishing a reference response against which the effects of introducing a culvert opening can later be evaluated. By excluding the culvert geometry in this initial stage, the dynamic response is governed primarily by the soil and track properties, allowing the influence of soil stiffness and damping to be evaluated without additional geometric disturbances.

In addition, the track model is used as a calibration platform, where selected soil parameters are adjusted to achieve agreement between the frequency response functions (FRFs) obtained from experimental measurements and those derived from the finite element model. The calibrated values, for the ballast and underlying and subgrade soil parts, are subsequently adopted as reference parameters in the more complex culvert model, ensuring consistency in material representation and improving the reliability of the overall numerical analysis.

The track model was established based on available design drawings and geotechnical data. The model has a total length of 40 m, providing a sufficiently large soil domain to ensure realistic wave propagation and track-soil interaction. The model was locked in translational motion at the bottom surface, representing the bedrock conditions at the bottom of the structure.

The model consists of four main structural components: the soil domain, ballast layer, sleepers and rail pads, and rails, as shown in Figure 12. The soil domain was represented using two distinct layers: subgrade soil and underlying soil. Initial material properties for these layers were assigned based on estimated parameters reported in previous studies and available geotechnical information. These parameters served as preliminary input values and were later refined through the calibration process using experimental frequency response functions (FRFs). Material damping was introduced through assigned damping ratios for the soil layers to account for energy dissipation during dynamic loading.

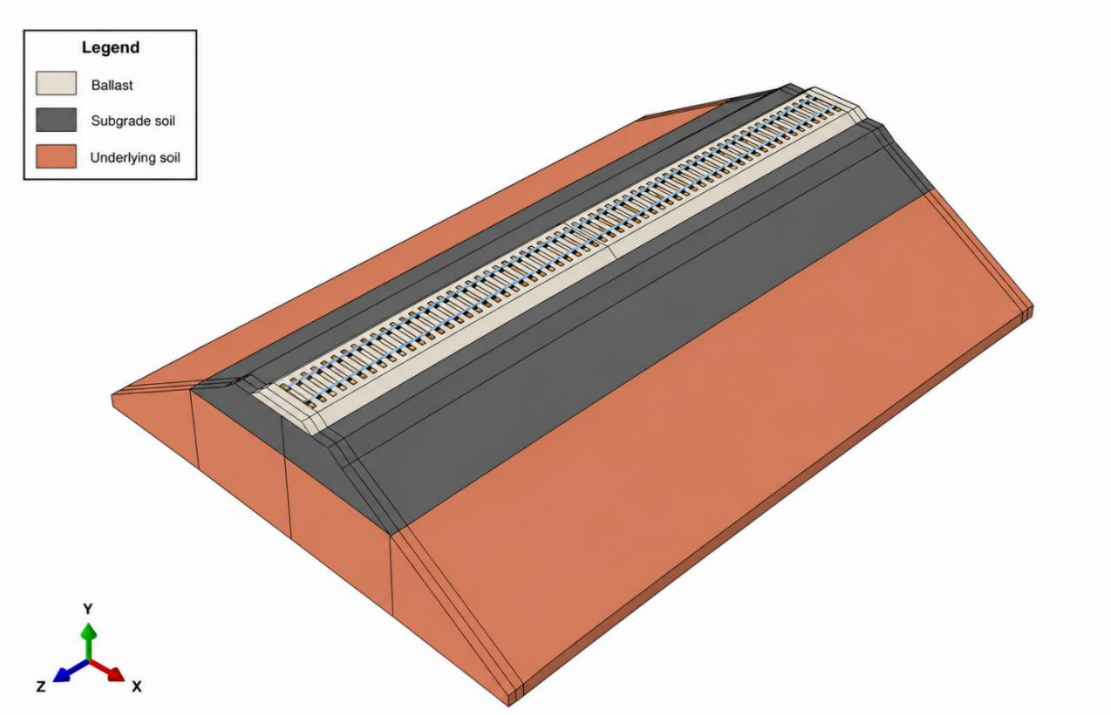


Figure 12: Three-dimensional finite element model of the benchmark track–soil system developed in Brigade Plus, illustrating the ballast, subgrade soil, and underlying soil layers included in the numerical analysis.

The soil domain was modeled in Brigade Plus using two homogeneous solid parts. The ballast layer was modeled as a separate solid part, while the subgrade and underlying soil were modeled as a second solid part. The latter was subsequently partitioned to define the different soil layers, to which individual material properties and soil profiles were assigned. Similarly, the sleepers and pads were modeled as homogenous solids, where the material properties and spacing were in accordance with standard *Trafikverket* specifications. The steel rails were modeled as wire beam elements with an I-beam profile resembling standard rail UIC60 steel beam profile. The material properties assigned to the rail substructure including the rails, sleepers, and pads in the numerical model are summarized in Table 3.

Table 3: Material properties assigned to the rails, sleepers, and rail pads in the finite element track model.

Type of part	Parameter	Value
Rails	Elastic modulus (MPa)	210000
	Density (Kg/m^3)	7850
	Poisson ratio	0.3
Pads	Elastic modulus (MPa)	20000
	Density (Kg/m^3)	1000
	Poisson ratio	0.2
Sleepers	Elastic modulus (MPa)	20000
	Density (Kg/m^3)	500
	Poisson ratio	0.2

The interaction between the different structural components of the track model was defined to realistically represent the load transfer mechanisms between the rails, pads, sleepers, ballast, and soil layers. The sleepers were connected to the ballast using an embedded region constraint, where the sleepers were defined as the embedded region and the ballast as the host material. This approach ensures compatible displacement between the two components. The ballast and soil domain were connected using tie constraints where main soil was defined as the master surface and ballast as the slave surface.

The interaction between the rails and the pads was represented using discrete spring elements, where the rail node was assigned as the master node and the pad center node as the slave node. The springs introduce directional stiffness in the longitudinal, lateral, and vertical directions, enabling a realistic representation of the rail–pad system. The spring stiffness values assigned in the X-, Y-, and Z-directions are summarized in Table 4. Furthermore, tie constraints were established between each pad center node and the corresponding pad surface to ensure displacement compatibility at the contact interface.

Table 4: Spring directional stiffness connecting the rail-pad elements.

Direction	Symbol	Stiffness value (N/m)
Longitudinal (X)	D11	2500000
Lateral (Y)	D22	200000000
Vertical (Z)	D33	2500000

To reduce unrealistic reflection of waves at the model boundaries of the numerical domain, additional soil extensions and infinite element material properties were introduced at both longitudinal ends of the track model, as illustrated in Figure 12. Two separate parts of “added soil”, each with thickness of 0.5 m and the same properties as the main soil part, were included to provide a transition zone between the main model and the boundaries.

Beyond these extensions, “infinite layer” parts, similarly each with thickness of 0.5 m and the same properties as the main soil part, were added to model the track extension, as illustrated in Figure 13. The element type in the “infinite layer” parts was later transformed to infinite elements by editing the input file of the Brigade model. This was performed using a MATLAB script, presented in Appendix B, that converted C3D20 mesh elements in the “infinite layer” parts to CIN3D12R mesh elements. The CIN3D12R elements are 12-node infinite elements used to represent an unbounded soil domain.

This infinite element application was done to simulate a continuous soil medium allowing waves generated by dynamic loading to propagate out of the model without reflecting into the analysis domain. In this way, the infinite material properties represent the physical continuation of the track and surrounding ground beyond the limits of the numerical model.

The interaction between the main soil domain and the added soil layers was defined using a tie constraint, where the added soil acted as the master surface and the main soil as the slave. A similar

tie interaction was applied between the added soil and the infinite layers, with the infinite layers defined as the master surface.

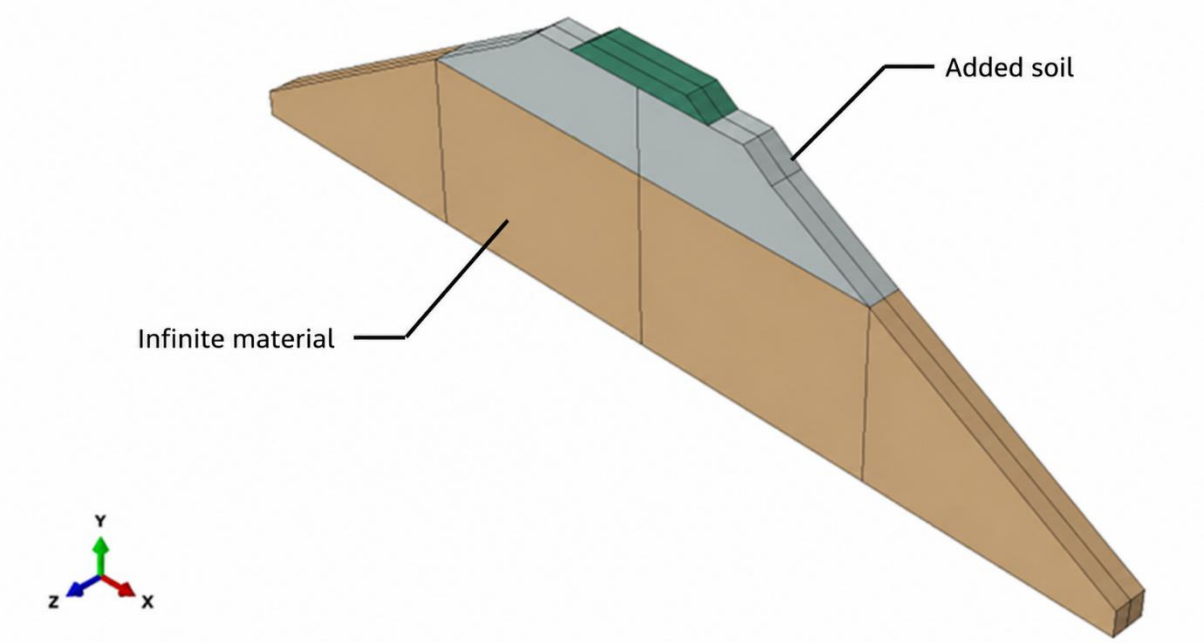


Figure 13: Illustration of the added soil and infinite material layers applied at the longitudinal boundaries of the numerical model to reduce artificial wave reflections during the dynamic analysis.

5.2 Orthotropic Culvert Bridge Model

The culvert model was developed as a continuation of the previously established track model, where the same modelling assumptions, boundary conditions, and calibrated material parameters were retained to ensure consistency between the two numerical models. The purpose of extending the model was to introduce the culvert opening within the soil domain and investigate how its presence influences the soil–structure interaction (SSI) behavior compared to the benchmark track model.

In addition to the structural components included in the track model, two additional soil components were introduced to represent the soil conditions surrounding the culvert: the engineered backfill surrounding the culvert and the underpinning soil located beneath it. The complete geometry of the culvert model, including the railway track, culvert opening, and surrounding soil layers, is illustrated in Figure 14.

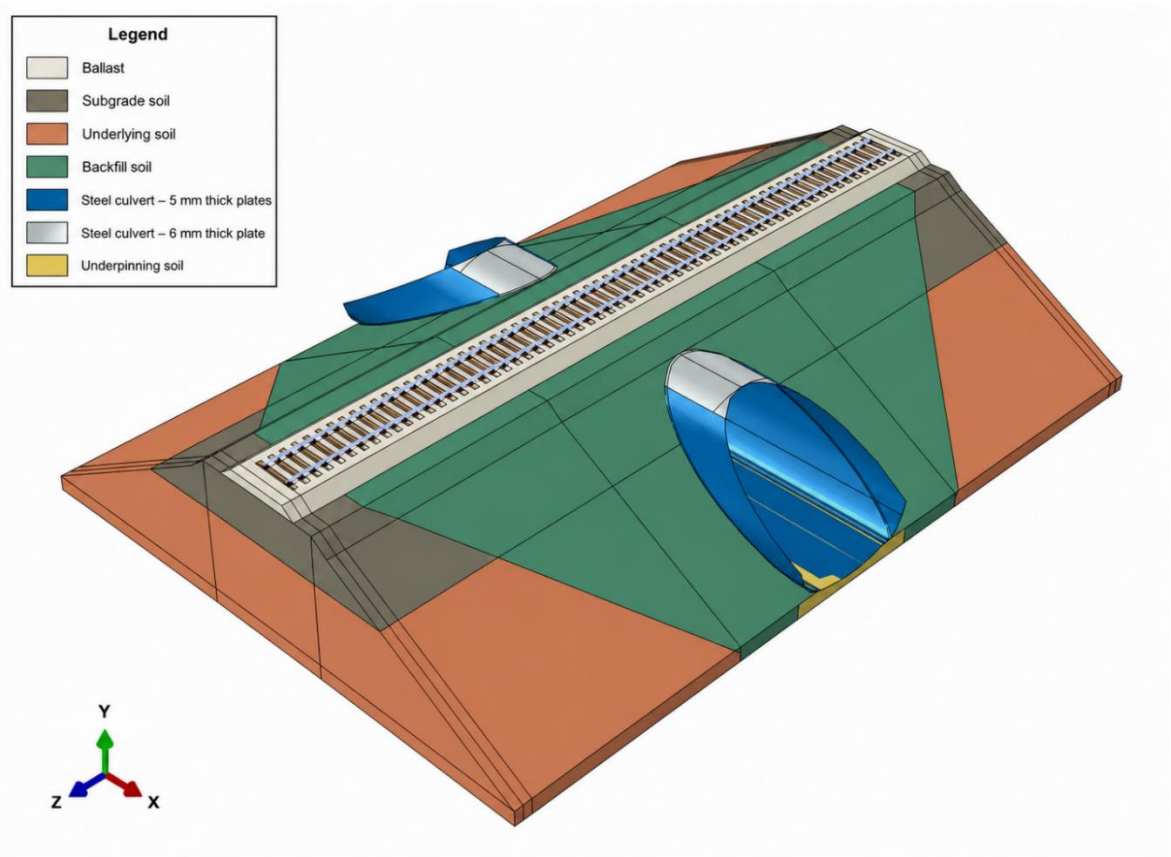


Figure 14: Three-dimensional finite element model of the culvert–soil system developed in Brigade Plus, illustrating the railway track, ballast, subgrade soil, underlying soil, backfill soil, underpinning soil, and the equivalent orthotropic steel culvert included in the numerical analysis.

The culvert opening was created by first removing the corresponding geometry from the backfill soil domain. The culvert cross-section was then defined based on available design drawings, as shown in Figure 5. Since explicitly modelling the detailed corrugated steel geometry would significantly increase the computational cost of the dynamic analyses, a simplified but sufficiently accurate modelling approach was adopted. Instead of modelling the actual corrugated profile, the steel culvert was represented using equivalent orthotropic shell elements with a rectangular cross-section, as illustrated in Figure 15. The equivalent orthotropic shell approach adopted in this study has been widely used in previous research on soil–steel structures [1]. The method enables the global stiffness behavior of the corrugated steel structure to be captured while maintaining computational efficiency. The equivalent orthotropic shell properties were derived to represent the stiffness characteristics of the corrugated profile in both the longitudinal and circumferential directions.

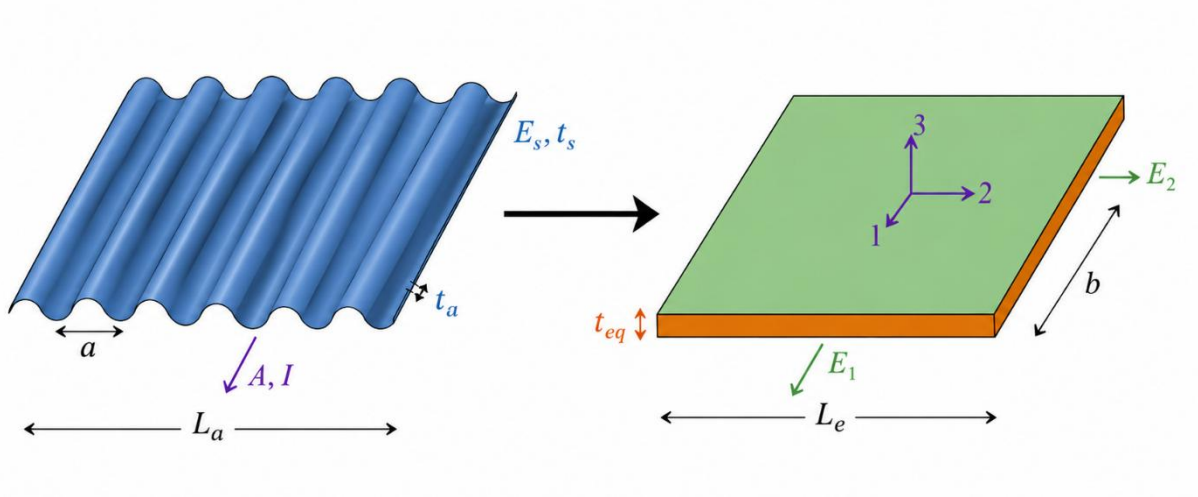


Figure 15: Illustration of the equivalent orthotropic shell modelling approach adopted for the corrugated steel culvert, where the corrugated steel profile is represented by an equivalent flat orthotropic shell with equivalent thickness and stiffness properties.

Using Equations (20-26) [1], the equivalent properties of the orthotropic shell elements were determined, including the equivalent thickness (t_{eq}), the circumferential and longitudinal Young's moduli (E_1 and E_2), the equivalent shear moduli in the respective planes (G_{13}, G_{23} and G_{12}), and the equivalent density of the orthotropic plate (ρ_{eq}).

$$E_1 = \frac{E_S A^{\frac{3}{2}}}{\sqrt{12\, I}} \tag{20}$$

$$E_2 = E_S \left(\frac{t_s}{t_{eq}}\right)^3 \tag{21}$$

$$t_{eq} = \frac{E_S\, A}{E_1} \tag{22}$$

$$\rho_{eq} = \frac{\rho_S\, A}{t_{eq}} \tag{23}$$

$$G_{13} = \frac{E_S\, A}{2(1 + \nu_S) t_{eq}} \tag{24}$$

$$G_{23} = \frac{E_S\, t_s}{2(1 + \nu_S) t_{eq}} \tag{25}$$

$$G_{13} = \frac{\sqrt{E_1\, E_2}}{2(1 + \nu_S)} \tag{26}$$

Here, E_S, ν_S, ρ_S and t_s donate the young's modulus, Poisson's ratio for steel, density and thickness of corrugated steel plate, respectively. A and I represent the cross-sectional area and the area of inertia per unit width of the corrugated profile. The values obtained of the equivalent parameters are presented below in Table 5.

Table 5: Equivalent orthotropic steel shell properties.

Property	**Symbol**	**Value**	
Corrugated Thickness	t_s (mm)	5	6
Elastic Modulus	E1 (MPa) longitudinal	83.93	144.01
	E2 (MPa) circumferential	18315.19	21913.93
Equivalent Thickness	t_{eq} (mm)	67.9	68.0
Equivalent Density	ρ ($\frac{Kg}{m^3}$)	684.639	819.163
Damping	ζ	0.01	0.01
Shear Modulus	G13(Mpa)	7044.31	8428.43
	G23 (Mpa)	5949.58	7122.62
	G12(Mpa)	476.87	683.26

After defining the orthotropic culvert shell and completing the geometric assembly of all components, a tie constraint was established between the outer surface of the culvert, defined as the master surface, and the inner surface of the backfill opening, defined as the slave surface. This ensured displacement compatibility and a proper structural connection between the culvert and the surrounding soil.

All remaining interaction definitions between the rails, pads, sleepers, ballast, and soil layers were retained from the previously established track model. Similarly, the same boundary treatment was adopted, where infinite elements were applied at the longitudinal boundaries of the ballast, subgrade soil, and underlying soil to reduce wave reflections and minimize their influence on the dynamic response of the system.

5.3 Model Mesh

The objective was to determine appropriate element size that provides a stable dynamic response, while keeping the computational cost at an acceptable level. The maximum acceptable mesh size was set to fulfill soil wave propagation criterion where there must be at least 8 elements per wavelength. The mesh was set to quadratic second order elements, so the effective element number is halved, where wavelength is represented by Eq (27) [15] as follows:

$$\Delta \leq \frac{v_p}{f_{max}} \tag{27}$$

Where Δ denotes the wavelength, v_p represents the wave velocity in the material, and f_{max} is the highest frequency of interest in the analysis. Based on this relation, a maximum element size, L_e, for a maximum frequency of 30 Hz and a conservative soil property estimation, was obtained for the soil parts based on Eq (28) as follows:

$$L_e = \sqrt{\frac{E}{2(1+\nu)\rho}} \cdot \frac{1}{4.f_{max}} = \sqrt{\frac{80.10^6}{2(1+0.2)1800}} \cdot \frac{1}{4.30} = 1.1\ m \tag{28}$$

After establishing the maximum allowable element size based on the wave propagation criterion, a mesh sensitivity analysis was performed to verify the stability and convergence of the numerical results. Modal analyses and response function (FRFs) were evaluated using several monitoring nodes on the track and culvert models to capture the dynamic response. This step was performed before applying infinite material properties to the models. Different mesh sizes were tested progressively for the ballast and soil parts in the track model, and for the steel culvert part in the culvert model, where the element size was reduced stepwise until the variation in response between successive simulations became negligible, indicating convergence at that respective node.

5.3.1 Track model

The mesh convergence study was performed by comparing the frequency response functions (FRFs) obtained from numerical hammer impact simulations using progressively refined meshes. The convergence assessment was based on the stability of the FRF curves, where further mesh refinement resulted in negligible changes in the predicted dynamic response. The results showed that the ballast response converged at an element size of 0.3 m, while no significant changes in the FRFs of the subgrade and underlying soil layers were observed for element sizes smaller than 0.8 m. The mesh convergence plots are presented in Appendix A, and the final meshed model is shown in Figure 16. The element type and mesh size assigned to the Added soil and Infinite layer parts were selected to match those adopted for the corresponding soil layers.

Mesh sensitivity of the sleepers and rails was not considered necessary. These parts represent comparatively stiff structural elements whose local discretization had negligible influence on the global soil-dominated dynamic behavior. The adopted mesh size and element type for the track model are summarized in Table 6.

Table 6: Adopted element mesh size and type for the track model.

Model component	Element type	Element size (m)
Ballast	C3D20R	0.3
Soil (Underlying & Subgrade)	C3D10	0.8
Sleepers	C3D8R	0.2 length 0.1 depth
Rails	B23	0.2

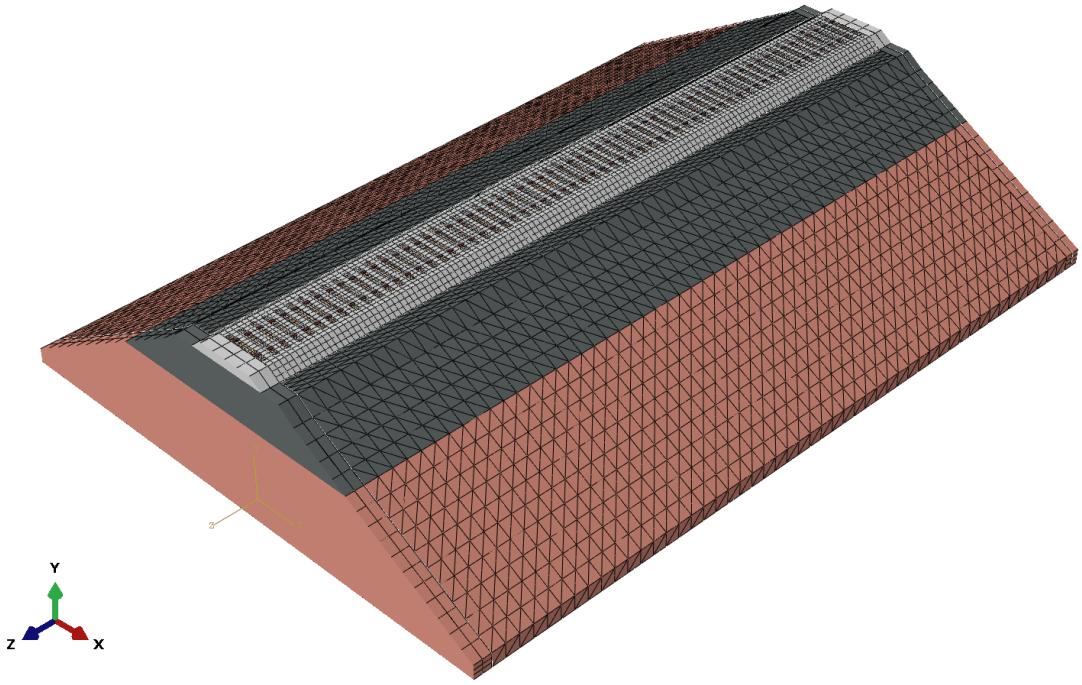


Figure 16: Meshed finite element model of the benchmark track–soil system including the added soil and infinite material layers at the longitudinal boundaries.

5.3.2 Culvert model

The mesh configuration established for the track model was used as the foundation for the culvert model to ensure consistency and comparability between the two configurations. The previously determined mesh sizes for the ballast and soil parts, and boundary regions were retained.

However, due to the geometric discontinuity introduced by the culvert opening and the expected concentration of stresses at the soil-structure interfaces, additional refinement was required in the zone of backfill and underpinning soils surrounding the steel culvert part.

A dedicated sensitivity analysis was conducted for the culvert shell part that indicated that this component achieved stable dynamic response at an element size of 0.4 m. The mesh convergence plot is shown in Appendix A, and the finalized meshed model is shown in Figure 17. A local mesh edge size for the backfill, and the underpinning soil around the culvert was adopted to be 0.6 m to ensure more compatible discretization across culvert-soil interface and improved numerical stability in the interaction region. The adopted element sizes for the culvert model are summarized in Table 7.

Table 7: Adopted element mesh size and type for the culvert model.

Modal component	Element type	Element size (m)
Ballast	C3D20R	0.3
Soil (Underlying & Subgrade)	C3D10	0.8
Sleepers	C3D8R	0.2 length 0.1 depth
Rails	B23	0.2
Added soil	C3D20R	0.8
Infinite layer	C3D20R	0.8
Backfill soil	C3D10	0.6
Underpinning soil	C3D10	0.8
Culvert shell	S8R	0.4

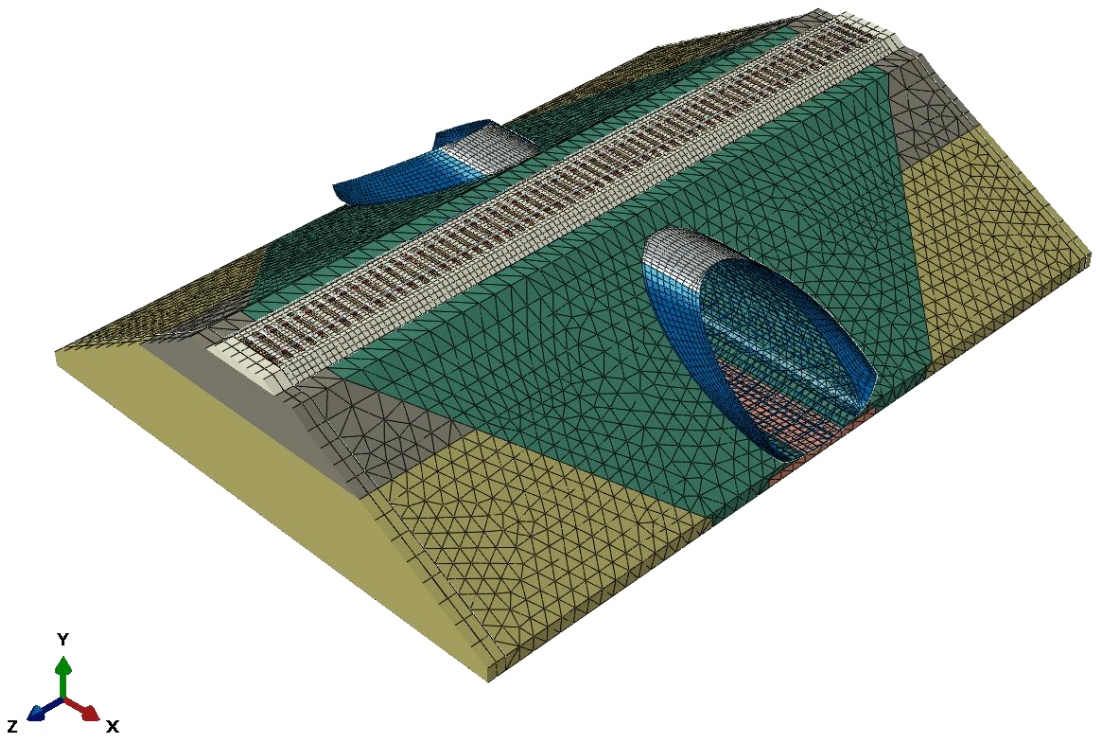


Figure 17: Meshed finite element model of the culvert–soil system including the added soil and infinite material layers at the longitudinal boundaries.

5.4 Calibrated Models

After obtaining the frequency response functions (FRFs) from the field measurements performed on the Åre culvert and the adjacent track section, the material parameters of the soil layers used in the numerical models were calibrated. In particular, the elastic modulus and damping ratios of the soil components were adjusted to achieve agreement between the numerical and experimental responses.

The calibration procedure was based on comparing the FRFs obtained from the finite element models with those derived from the hammer impact tests conducted in the field. In the numerical models, the FRFs were generated by applying harmonic loads at the same excitation locations used during the experimental hammer tests. A harmonic steady load step was created in Brigade and a vertical unit load of –1 kN was applied in the numerical models at each corresponding impact location to simulate the experimental loading conditions. The dynamic responses of magnitude of vertical acceleration were then extracted at the respective sensor locations to generate numerical FRFs.

The experimental FRFs were calculated from the measured hammer force and acceleration responses according to Eq (15) presented in Section 2.4. The numerical and experimental FRFs were subsequently compared by evaluating the dominant frequency peaks, peak amplitudes, and the overall shape of the response curves. Based on this comparison, the soil stiffness and damping parameters were manually iteratively adjusted until reasonable agreement between the numerical and experimental responses was achieved. The focus was to achieve a relatively good match between the FRFs in the first structural mode peak, in frequency and amplitude.

5.4.1 Track model

The calibration process was performed in a stepwise manner. First, the track model was calibrated, as it represents the reference benchmark model without the influence of the culvert opening and surrounding backfill. In this stage, the elastic modulus and damping ratios of the subgrade and underlying soil layers, in addition to the depths of these soil layers were iteratively adjusted until the numerical FRFs showed good agreement with the experimentally measured FRFs. The experimental FRFs used for this calibration were obtained from sensors A8, A9, A11, and A13 located on the track approximately 20 m from the culvert midspan, as shown in Figure 9.

The comparison between the numerical and experimental FRFs obtained from sensors A8, A9, A11, and A13 for impact location P7 are presented in Figures 18-21. Overall, the calibrated track model captures the dominant resonance behavior observed in the field measurements, where the first frequency peak is consistently reproduced around 19 Hz for all sensor locations. The numerical model also shows reasonable agreement with the measured FRF in terms of the amplitude shown of the first peak.

Some discrepancies are observed at higher frequencies, where the experimental FRFs exhibit additional local peaks and increased response amplitudes compared to numerical results. These differences are likely related to simplifications adopted in the numerical model, including the assumption of homogeneous soil layers, idealized boundary conditions, and simplified representation

of track components and damping behavior. Nevertheless, the calibrated model reproduces the overall dynamic characteristics of the track–soil system with sufficient accuracy and was therefore considered suitable for further calibration and development of the complete culvert model.

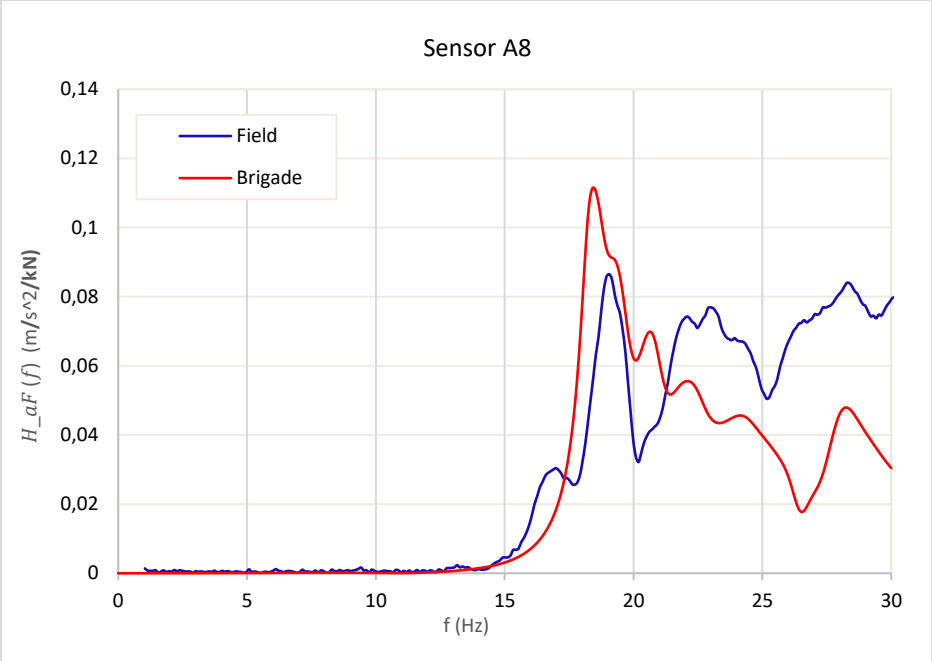


Figure 18: Comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated track model for sensors A8 subjected to hammer impact loading at location P7.

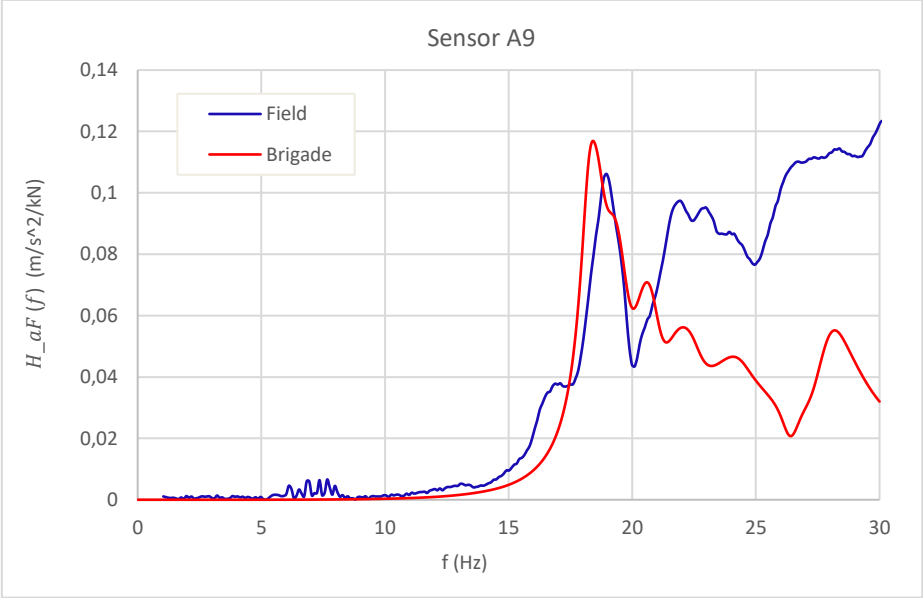


Figure 19: Comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated track model for sensor A9 subjected to hammer impact loading at location P7.

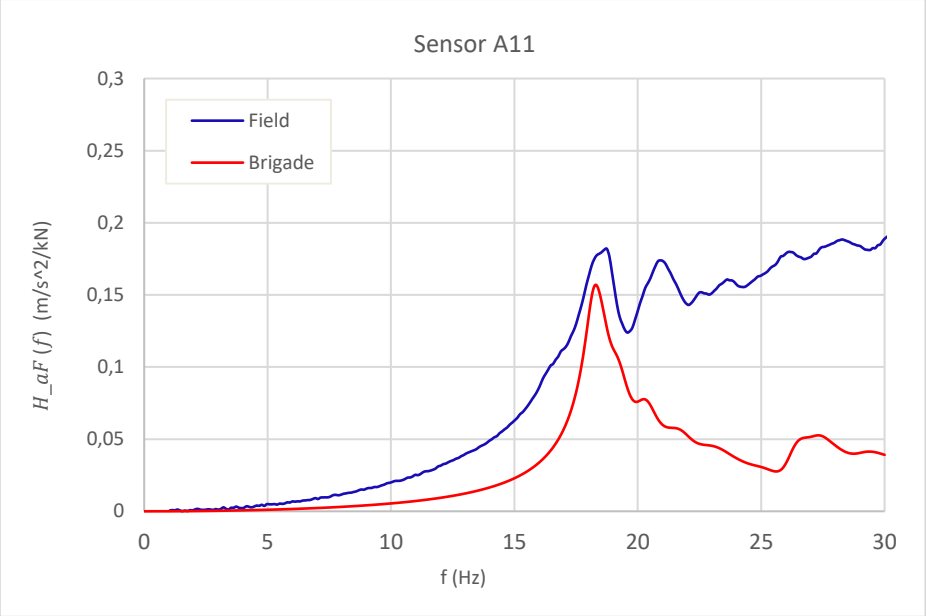


Figure 20: Comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated track model for sensor A11 subjected to hammer impact loading at location P7.

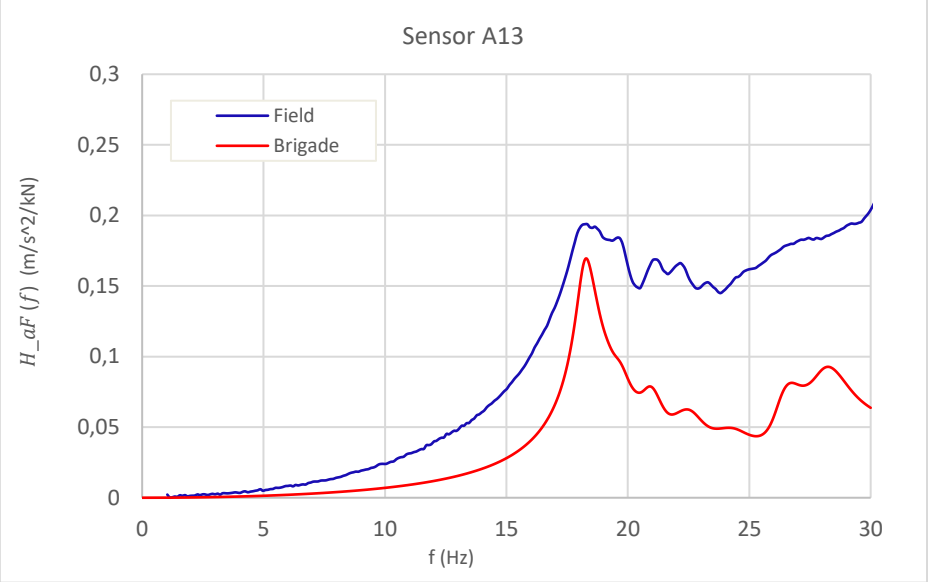


Figure 21: Comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated track model for sensor A13 subjected to hammer impact loading at location P7.

The calibrated material parameters obtained for the track model are summarized in Table 8. These values represent the soil properties that provided the best agreement between the numerical and experimental frequency response functions (FRFs).

Table 8: Calibrated material properties of the track model.

Type of soil	Parameter	Value
Ballast	Elastic modulus (MPa)	200
	Density (Kg/m^3)	1800
	Poisson ratio	0.2
	Damping ratio	0.04
Subgrade soil	Elastic modulus (MPa)	85
	Density (Kg/m^3)	2000
	Poisson ratio	0.2
	Damping ratio	0.04
Underlying soil	Elastic modulus (MPa)	2000
	Density (Kg/m^3)	2100
	Poisson ratio	0.2
	Damping ratio	0.06

5.4.2 Culvert model

Once reasonable agreement was achieved for the track model, the calibrated soil parameters were adopted as reference values for the culvert model. This approach ensured that the dynamic properties of the soil were consistently represented before introducing the additional soil layers required for the culvert model, namely the backfill soil surrounding the culvert and the underpinning soil beneath it.

Subsequently, the culvert model was calibrated by adjusting the elastic modulus and damping ratios of engineered backfill and underpinning soil layers. During this stage, the parameters of the backfill and underpinning soil were manually iteratively modified while keeping the previously calibrated subgrade and underlying soil properties obtained from track model calibration step.

For the impact loading at location P1-2, the comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated culvert model is presented in Figures 22-24 for sensor A1 located inside the culvert, sensor A3 positioned on the ballast above the culvert, and sensor A4 positioned on the sleeper above the culvert. The locations of the sensors and impact points are illustrated in Figures 9 & 10.

Overall, the numerical model reproduces the dominant dynamic characteristics observed in the field measurements with reasonable agreement. The calibrated model captures the primary resonance peak within approximately 20 Hz for all investigated sensor locations. Good agreement is observed for sensor A1, where both the resonance frequency and the general response trend are well represented. Similar agreement is obtained for sensors A3 and A4, although some differences in response amplitudes are observed at frequencies above the dominant resonance peak. Nevertheless, the numerical model captures the overall shape and behavior of the experimental FRFs, indicating that the calibrated material parameters provide a realistic representation of the dynamic soil–steel interaction behavior of the culvert system.

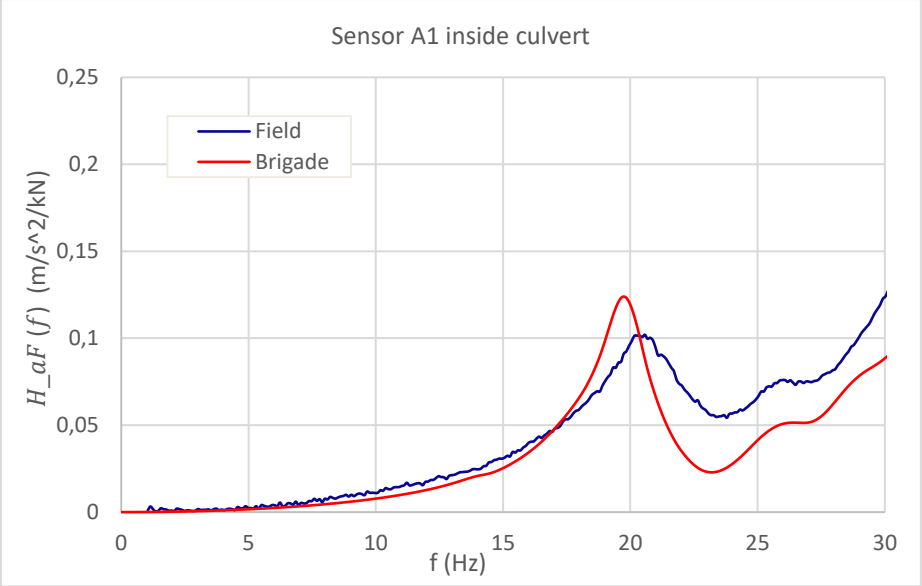


Figure 22: Comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated culvert model for sensor A1 inside the culvert, subjected to hammer impact loading at location P1-2.

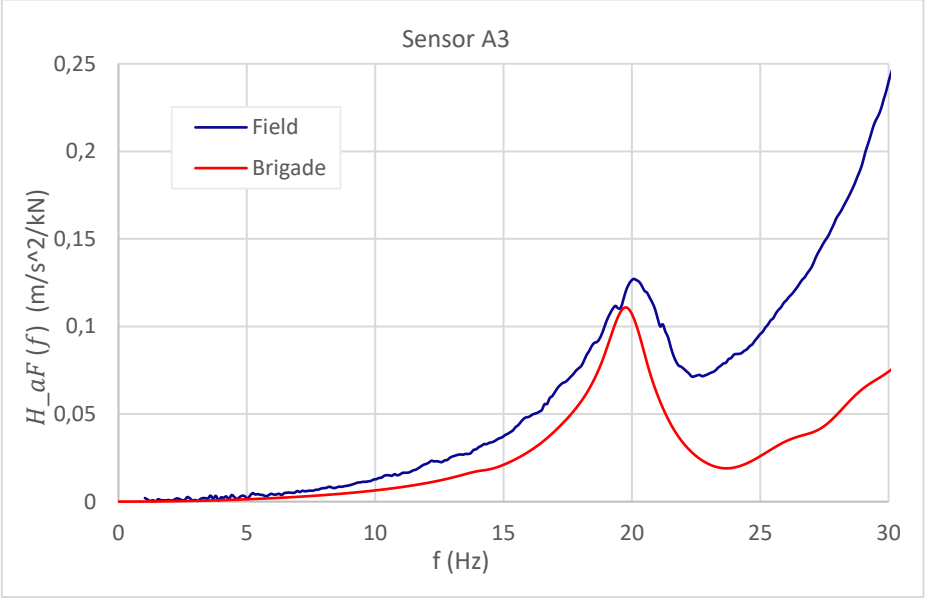


Figure 23: Comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated culvert model for sensor A3 subjected to hammer impact loading at location P1-2.

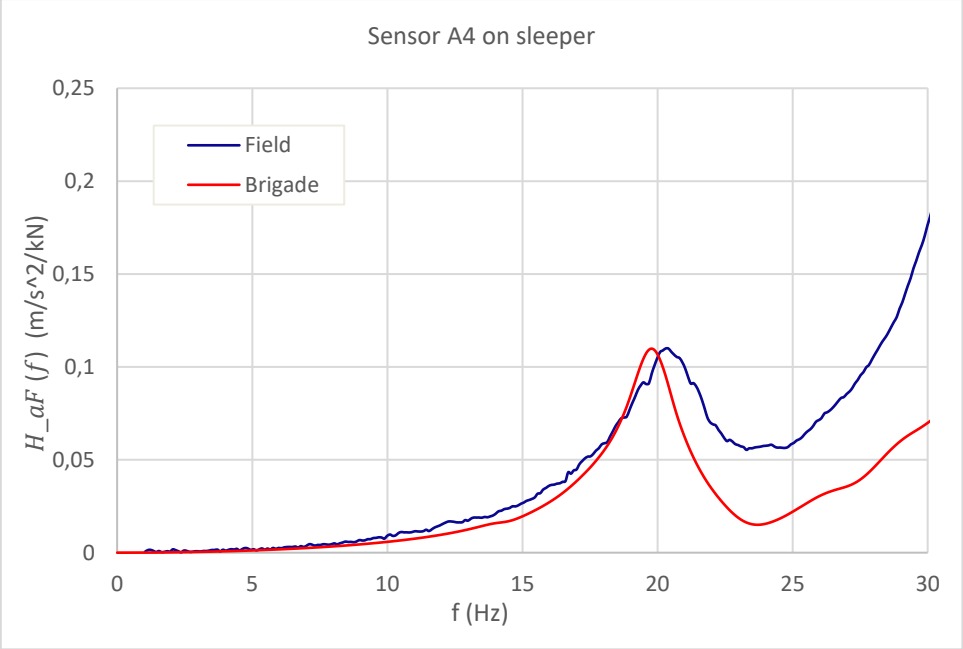


Figure 24: Comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated culvert model for sensor A4 subjected to hammer impact loading at location P1-2

For the impact loading at location P0-1 inside the culvert, the comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated culvert model is presented in Figures 25-27 for sensor A1 located inside the culvert, sensor A3 positioned on the ballast above the culvert, and sensor A4 positioned on the sleeper above the culvert. The locations of the sensors and impact points are illustrated in Figure 10.

Similar to the results obtained for impact location P1-2, the calibrated model successfully reproduces the first peak behavior and the general response trend observed in the field measurements. The primary resonance peak is captured within the same frequency range for all sensor locations, demonstrating consistency in the dynamic response of the numerical model. However, the experimentally measured FRFs exhibit slightly higher response amplitudes and additional local variations at higher frequencies compared to the numerical results.

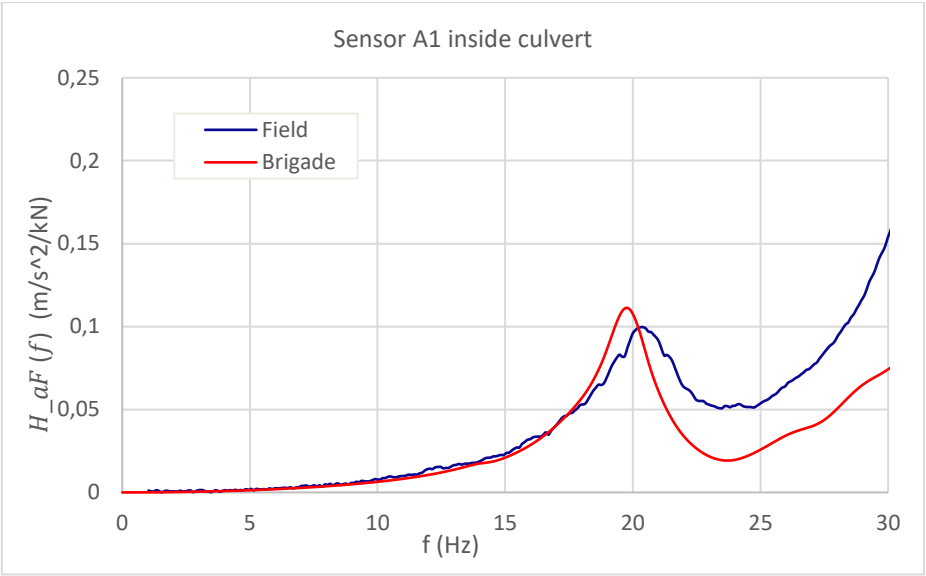


Figure 25: Comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated culvert model for sensor A1 inside the culvert, subjected to hammer impact loading at location P0-1.

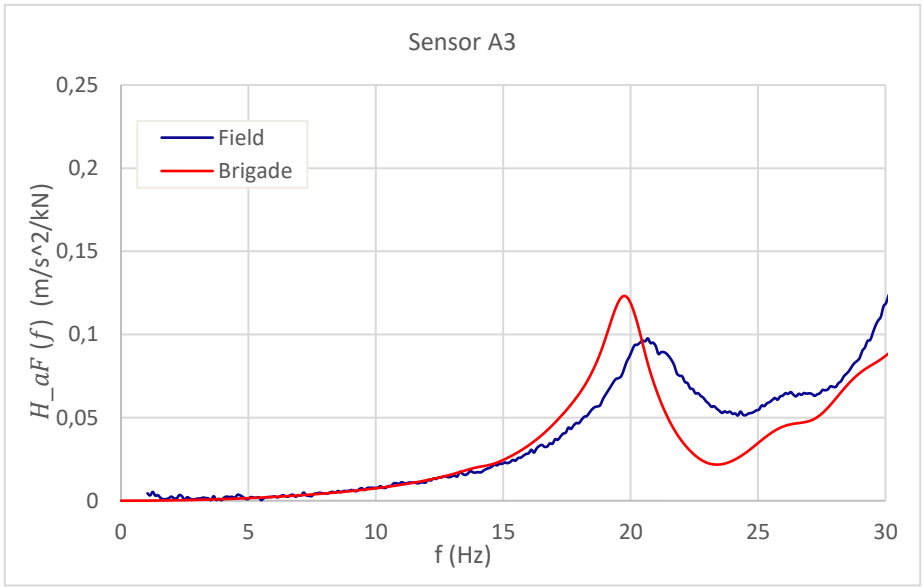


Figure 26: Comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated culvert model for sensor A3 subjected to hammer impact loading at location P0-1.

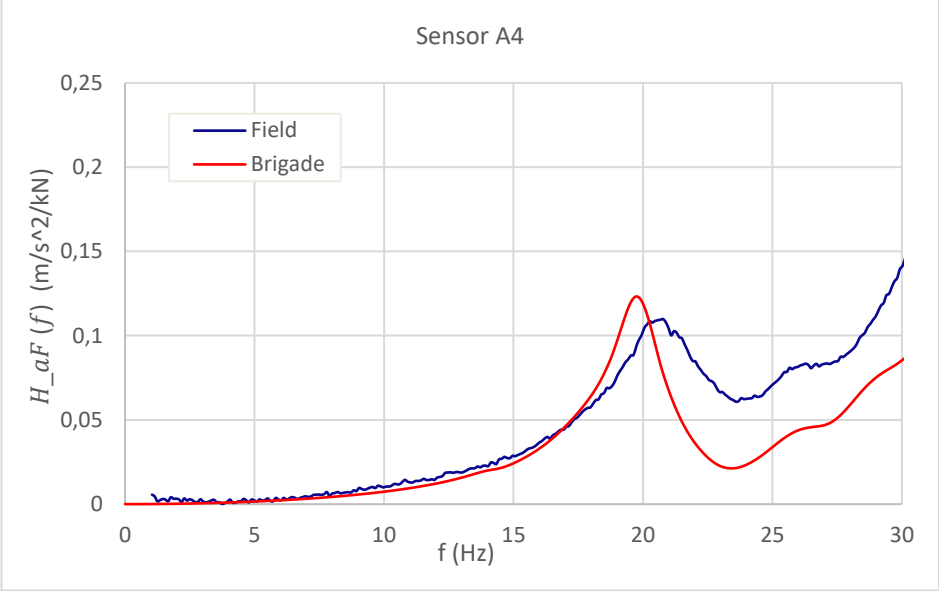


Figure 27: Comparison between the experimentally measured FRFs and the numerically obtained FRFs from the calibrated culvert model for sensor A4 subjected to hammer impact loading at location P0-1.

The calibrated material parameters obtained for the culvert model are summarized in Table 9 as follows:

Table 9: Calibrated material properties of the culvert model.

Soil layer	**Young's modulus E(MPa)**	**Density ρ(kg/m³)**	**Poisson's ratio ν**	**Damping ratio ξ(%)**
Backfill soil	600	1800	0.3	0.1
Underpinning soil	500	1800	0.3	0.04

After calibration of both the track and culvert models, Rayleigh wave velocity of the subgrade and backfill soil layers were calculated based on Eq (4-6). The corresponding Rayleigh wave velocity of the subgrade soil was estimated around 435 $\frac{km}{h}$, whereas the wave velocity of the backfill soil was estimated around 1194 $\frac{km}{h}$, as shown in Table 10 as follows:

Table 10: Estimated Rayleigh soil velocities based on calibrated material properties.

Soil layer	V_s $(\frac{m}{s})$	V_R $(\frac{m}{s})$	V_R $(\frac{km}{h})$
Subgrade	133	121	435
Backfill	358	332	1194

The resulting mode shape corresponding to the first natural frequency around 20 Hz is presented in Figure 28. As shown in the figure, the first dominant vibration mode corresponds to the first vertical bending mode of the culvert structure. This mode shape represents the fundamental dynamic behavior of the soil–steel culvert bridge, where the culvert undergoes a global vertical deformation while interacting with the surrounding soil. The identification of this mode is important, as it governs the dynamic response of the structure within the frequency range observed during the field measurements and the subsequent calibration process.



Figure 28: First vertical bending mode of the calibrated culvert model obtained from the finite element modal analysis, representing the fundamental vibration mode of the soil–steel culvert bridge.

5.5 Model Validation using Passing Trains

To further validate the calibrated numerical models, an additional step was carried out by comparing the dynamic responses obtained from the finite element simulations with the corresponding responses measured during monitored train passages at the site. Unlike the calibration procedure, which was primarily based on hammer impact testing and frequency response functions (FRFs), the train-passage analyses enabled the models to be evaluated under realistic operational loading conditions. This validation step was therefore used to assess the capability of the calibrated models to reproduce the actual dynamic behavior of the track–soil and culvert–soil systems subjected to moving train loads.

The validation procedure was performed for both the calibrated benchmark track model and the calibrated culvert model. The responses obtained from the numerical simulations were compared with experimentally measured acceleration and displacement signals recorded at selected sensor locations, matching the field sensor locations, during train passages. By evaluating the agreement between the numerical and experimental responses, the accuracy of the calibrated stiffness, damping, and interaction properties assigned to the different structural and soil components could be assessed. The monitored train passage included in the validation process is summarized in Table 2.

In the numerical simulations, the train passages were represented by moving point loads applied along the rails. The moving load configurations were generated using a MATLAB-based implementation developed in a previous similar PhD study conducted at KTH. The MATLAB code incorporated the axle arrangements, axle loads, axle spacing, and train velocities corresponding to the monitored train types and was used to generate time-dependent moving load sequences. These load histories were subsequently applied in the finite element analyses performed in Brigade Plus, allowing the numerical simulations to closely replicate the train loading conditions recorded during the field measurements.

Prior to the comparison between field data and FEM results, the recorded field data from the X62 Regina passing train was examined to understand the response of the physical bridge to current train loads. The vertical acceleration response at the culvert bridge section showed a relatively higher amplitude in Sensor A3 (midspan in the ballast) in comparison to Sensor A1 (midspan on the steel culvert crown), as shown in Figure 29. Indicating that the ballast at the same section is more sensitive to the train load than the corrugated steel culvert. Following, the data from the adjacent track section, shown in Figure 30, displayed acceleration amplitudes very similar to the ones seen in the track above the culvert section. This shows that the introduction of the bridge opening and steel culvert did not make the structure more sensitive to loading.

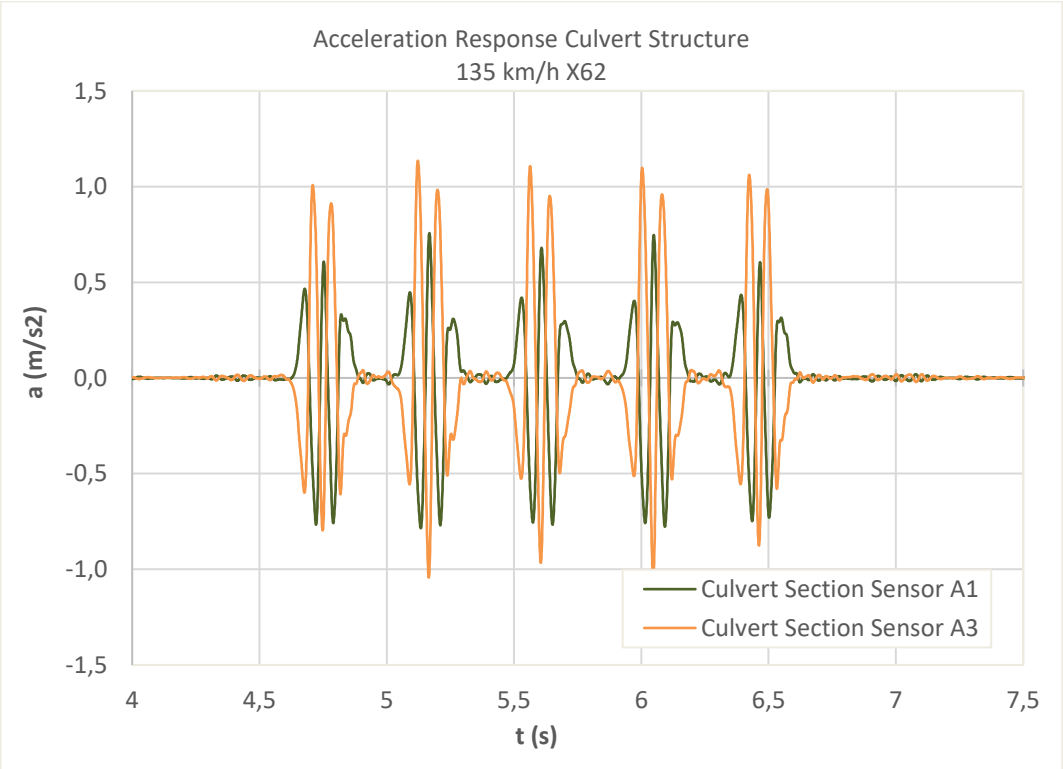


Figure 29: Vertical acceleration time history from recorded data during the passing of X62 Regina Train on Åre Culvert section.

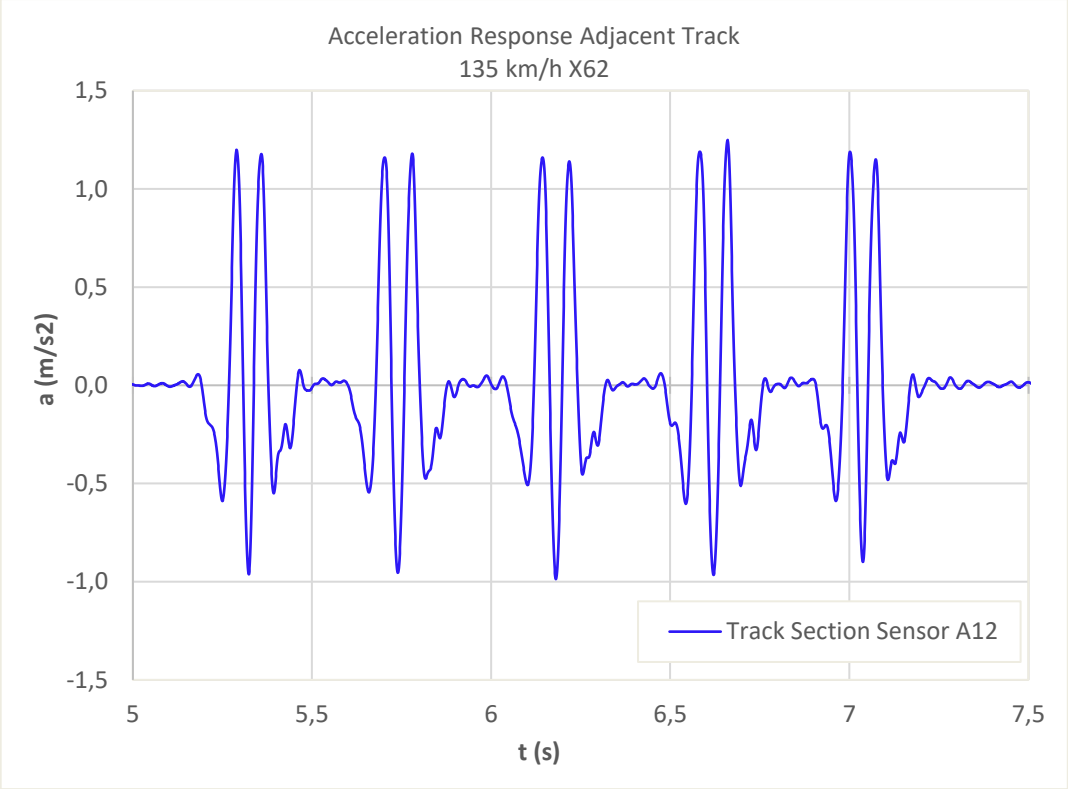


Figure 30: Vertical acceleration time history from recorded data during the passing of X62 Regina Train on the track section around 20 m away from Åre Culvert.

For the benchmark track model, the validation was performed by comparing the vertical acceleration responses obtained from the numerical simulations with the experimentally measured accelerations at sensor A12 located on the track. For the calibrated culvert model, the validation included comparisons of both acceleration and displacement responses. The acceleration responses were evaluated at sensor A1 located at the culvert midspan and sensor A3 positioned in the ballast above the culvert, while the displacement response was evaluated using the LVDT sensor D1 installed at the culvert midspan. This combination of measurements provided a comprehensive assessment of the ability of the numerical model to reproduce both the global structural response of the culvert and the dynamic behavior of the surrounding track–soil system. The locations of the sensors used in the validation process are illustrated in Figures 9 and 10.

The acceleration responses obtained from both the experimental measurements and the numerical simulations were post-processed using a Butterworth low-pass filter with a cut-off frequency of 30 Hz. The filtering was introduced to suppress high-frequency noise and local vibration effects associated with wheel–rail interaction, thereby emphasizing the global dynamic behavior of the track and culvert systems. This processing enabled a more reliable comparison between the measured and simulated responses with respect to the dominant vibration modes and resonance characteristics of the investigated structures.

The simulated responses were extracted at the same measurement locations as those used during the field monitoring to enable a direct comparison between the numerical and experimental results. Comparisons between the measured and simulated responses for the monitored X62 Regina train passage are presented in Figures 31–34.

Overall, the numerical models reproduce the general response patterns observed in the field measurements with reasonable agreement. For the benchmark track model, the simulated acceleration response at sensor A12 captures the timing and overall shape of the measured response during the train passage, see Figure 31. However, the numerical model underestimates the measured acceleration amplitudes, with the simulated peak values being approximately half of those recorded experimentally. A similar trend is observed for the culvert model, where the acceleration responses at sensors A1 and A3 exhibit lower amplitudes than those measured in the field, as shown in Figures 33 & 34.

A similar underestimation is observed for the displacement response amplitude at the culvert midspan, as shown in Figure 32. The numerical model captures the shape of the measured displacement response during the train passage reasonably but underestimates the amplitude by a factor of 2. The underestimation of amplitude in acceleration displacement may be attributed to modelling simplifications, uncertainties in damping properties, idealized boundary conditions, and limitations associated with representing the complex train–track–soil interaction using moving point loads. More specifically that the FE models are stiffer than the physical structures. Nevertheless, the overall agreement obtained for both acceleration and displacement responses demonstrates that the calibrated numerical models provide a relatively good representation of the dynamic behavior of the investigated systems, where a correction factor can be considered when examining amplitude, before the subsequent high-speed train analyses.

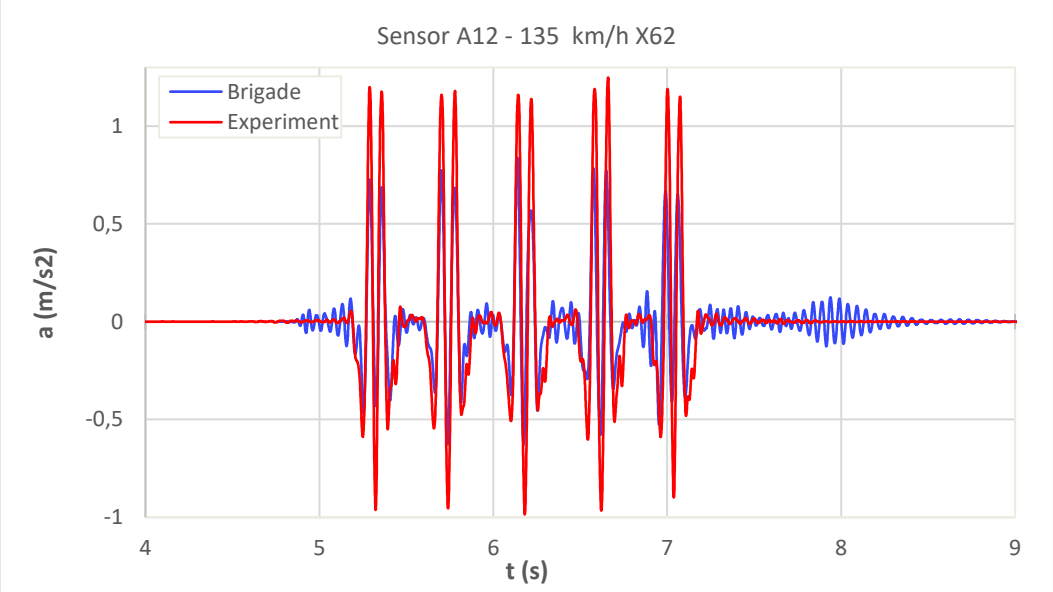


Figure 31: Comparison between the experimentally measured and numerically simulated vertical acceleration responses at sensor A12 for the X62 Regina train passage at a speed of 135 km/h.

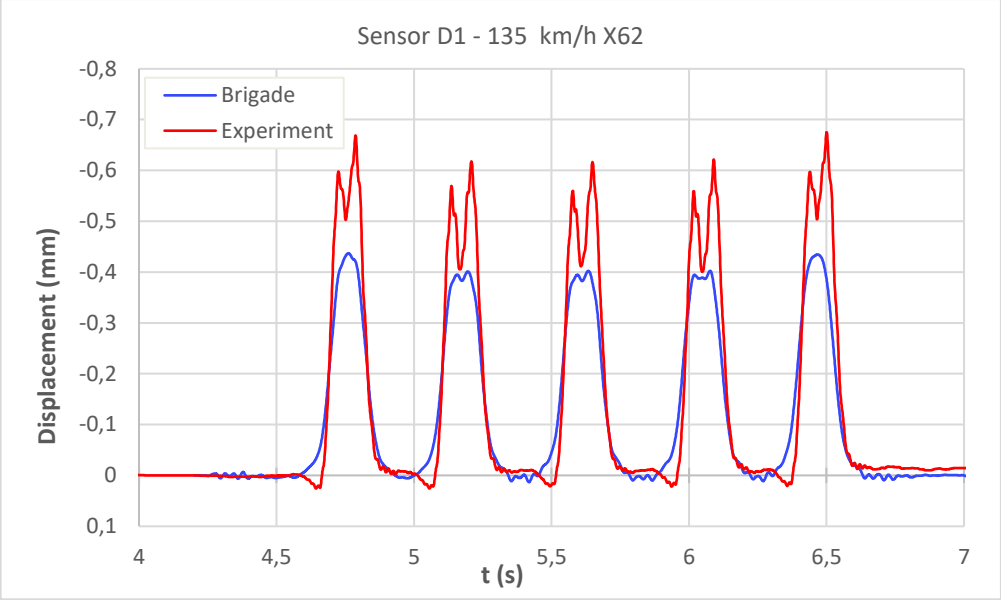


Figure 32: Comparison between the experimentally measured and numerically simulated vertical displacement responses at sensor D1 located at the culvert midspan for the passage of the X62 Regina train at a speed of 135 km/h.

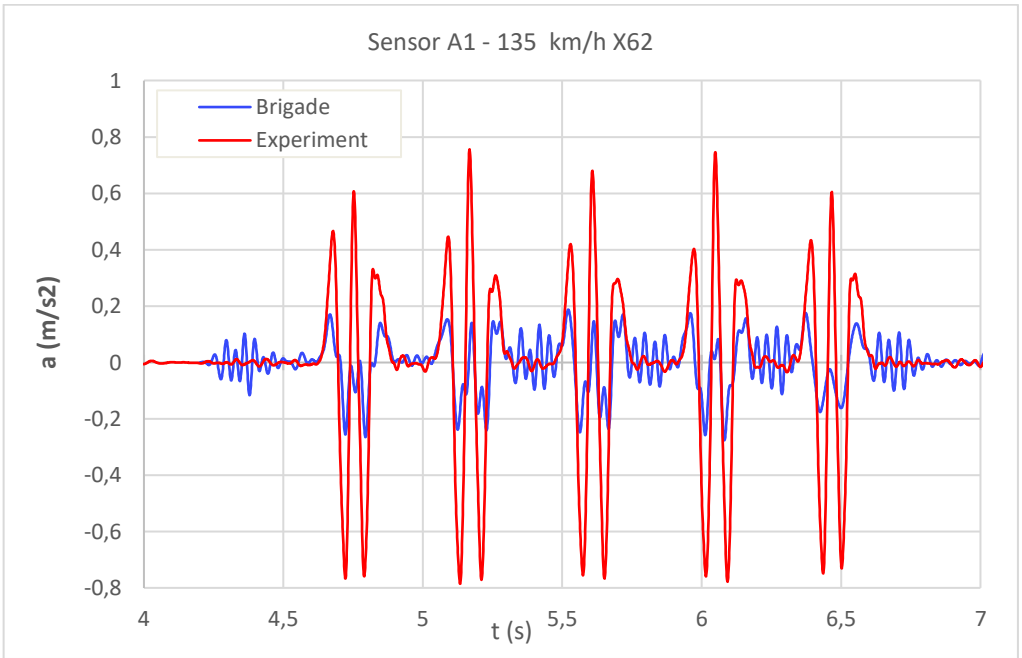


Figure 33: Comparison between the experimentally measured and numerically simulated vertical acceleration responses at sensor A1 located at the culvert midspan for the passage of the X62 Regina train at a speed of 135 km/h.

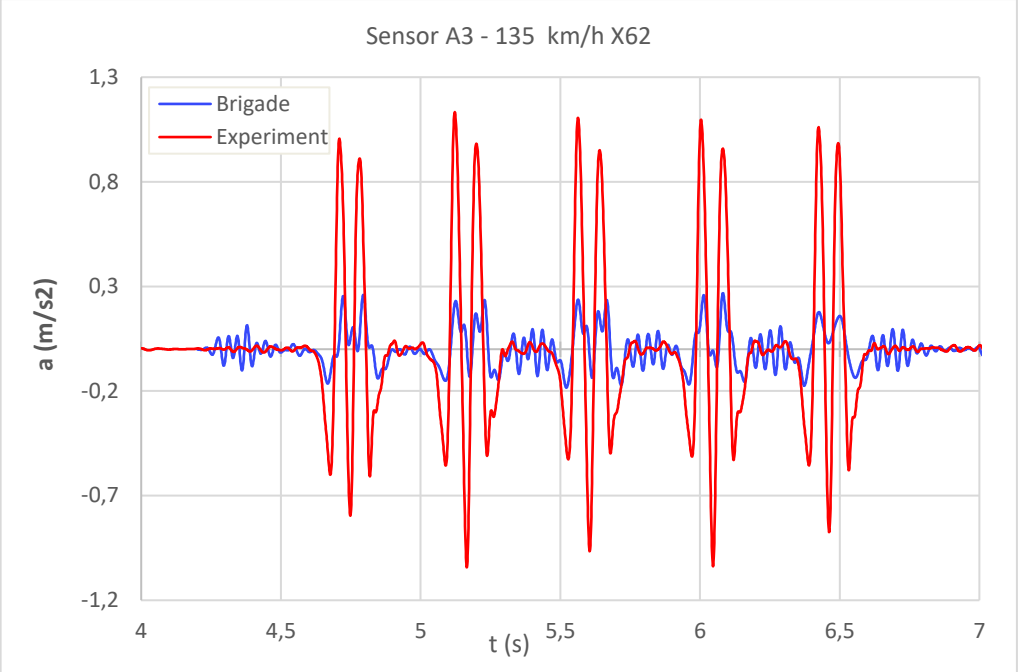


Figure 34: Comparison between the experimentally measured and numerically simulated vertical acceleration responses at sensor A3 for the passage of the X62 Regina train at a speed of 135 km/h.

6 High Speed Train Passage Analysis

The calibrated culvert and track models are used to simulate high-speed train passage and perform dynamic check on the soil-steel structure. The models were subjected to the design High-Speed Load Models (HSLM-A) specified by Eurocode as moving point loads. The moving load configurations were generated using a MATLAB-based implementation developed in a previous similar study in the Bridge PHD program at KTH. For each model, the train load case producing the largest acceleration response was selected and presented for train speeds ranging from 150 to 350 $\frac{\text{km}}{\text{h}}$. The acceleration responses were extracted from Abaqus, then filtered with cut-off frequencies of 30 Hz and 50 Hz to investigate the influence of the selected filtering frequency on the calculated peak response.

The comparison was performed between the calibrated culvert model and the calibrated adjacent track model. In the culvert model, the vertical acceleration response was evaluated at sensor A1, located on the culvert structure, and sensor A3, located in the ballast/backfill region above the culvert. These locations were selected because they represent the dynamic response of the soil–steel system at and above the culvert. The corresponding reference response was taken from sensor A11 in the adjacent track model, which represents the dynamic response of the ballasted track section away from the culvert influence. This comparison was used to assess whether the presence of the culvert opening introduces additional dynamic amplification relative to the adjacent track–soil system.

Figures 35 and 36 show the maximum vertical acceleration responses obtained for HSLM-A2 using 30 Hz and 50 Hz filtering, respectively. The HSLM-A2 load model produced the most critical acceleration response among the HSLM-A train configurations investigated. In both filtering cases, the track model at sensor A11 generally showed higher peak accelerations than the culvert model responses at sensors A1 and A3. This relatively large difference in amplitude between the track model and culvert model is likely due to over estimation of the stiffness of the backfill soil layer in the culvert model.

For the 30 Hz filtered results shown in Figure 35, the acceleration response of the track model at sensor A11 is consistently higher than the responses obtained from the culvert model at sensors A1 and A3 over the investigated speed range. The acceleration responses obtained at culvert sensors A1 and A3 are nearly identical throughout the investigated speed range. This does not match the trend observed from the recorded field data of currently operating trains as the ballast region and steel shell showed different responses. This indicates that the culvert structure and the surrounding backfill soil respond as a coupled system, which is likely a consequence of the full interaction assumed between the culvert shell and the surrounding soil in the numerical model.

When the filtering frequency is increased from 30 Hz to 50 Hz, as shown in Figure 36, the calculated peak acceleration values increase at higher speeds, particularly for the track model. This is expected because the higher cut-off frequency retains additional high-frequency components associated with the moving axle loads and local track response. However, the overall trend remains the same as the one displayed by the 30 Hz results.

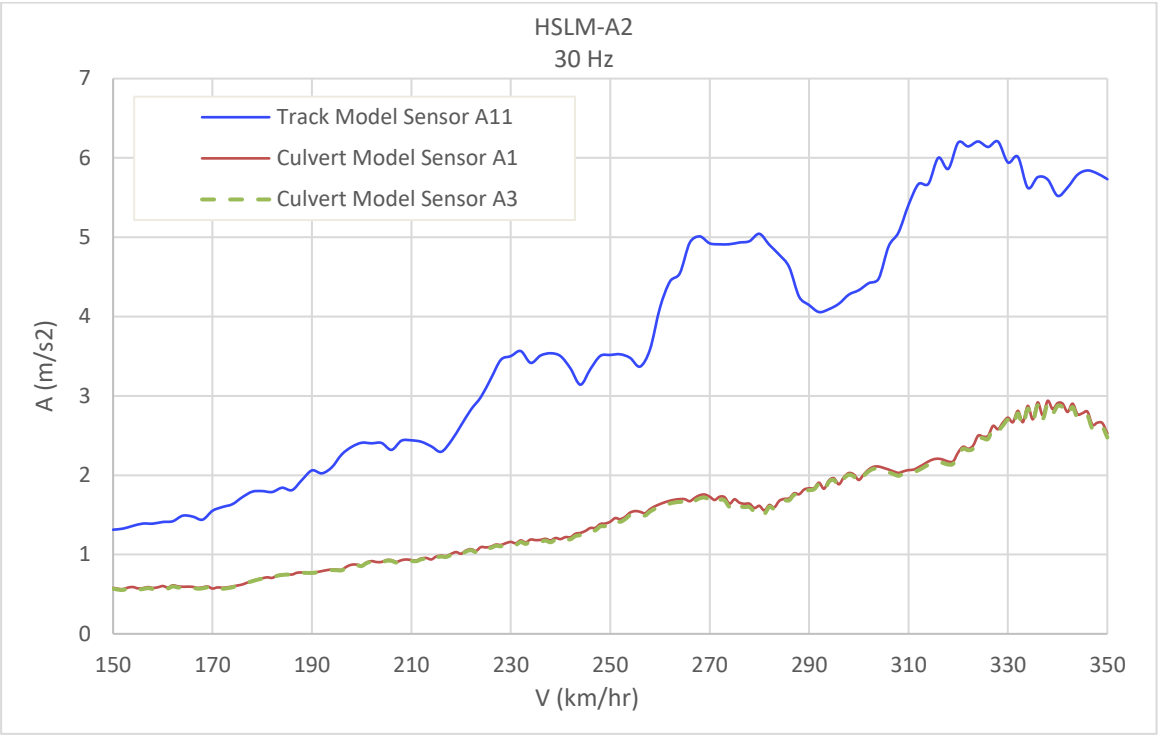


Figure 35: Maximum acceleration responses at sensor A1 and A3 (culvert model) and A11 (track model) during the HSLM-A2 train design loads, filtered at 30 Hz.

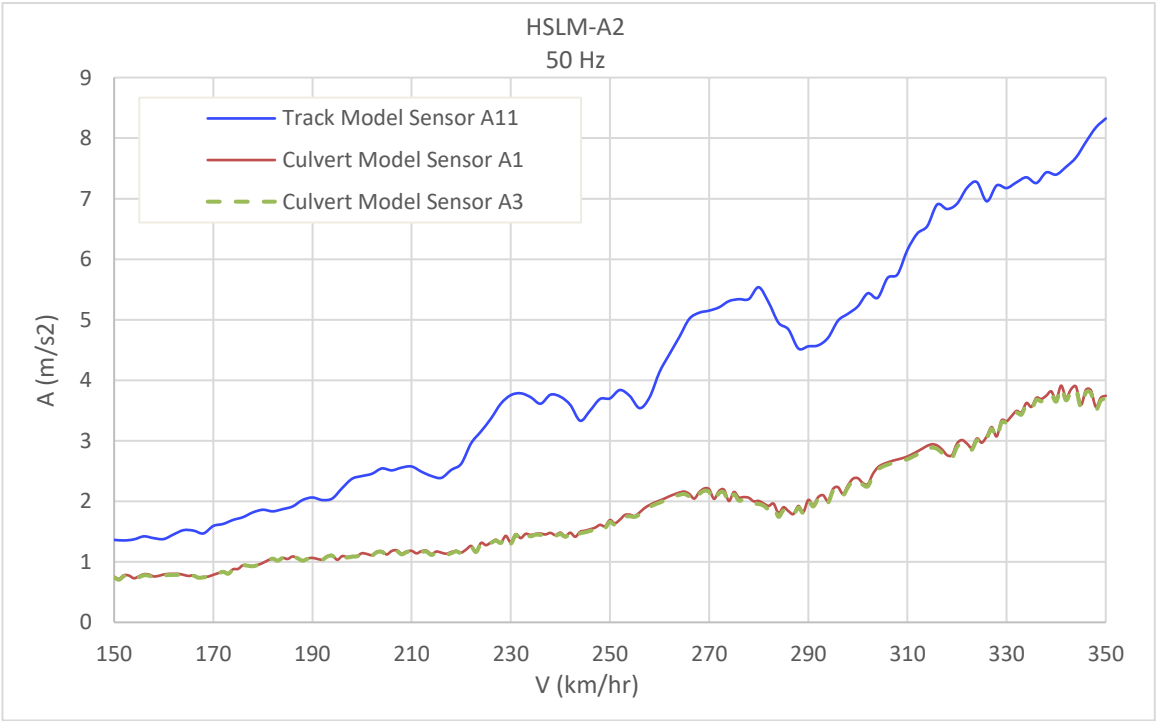


Figure 36: Maximum acceleration responses at sensor A1 and A3 (culvert model) and A11 (track model) during the HSLM-A2 train design loads, filtered at 50 Hz.

Additionally, the displacement responses obtained for the HSLM-A2 train loading are presented in Figure 37. Similarly to the acceleration results, the benchmark track model exhibits higher displacement levels than the calibrated culvert model throughout the investigated speed range. The displacement response at sensor A11 in the track model increases gradually with train speed, reaching a maximum value of approximately 0.85 mm at 350 km/h. In comparison, the displacement responses at culvert sensors A1 and A3 remain lower, with peak values of approximately 0.59 mm at the highest investigated speed. Furthermore, the displacement responses at culvert sensors A1 and A3 remain well below the commonly adopted serviceability limit of 2 mm for railway structures even with a correction factor of 2, at around 1.7 mm. These results suggest that the soil–steel culvert maintains satisfactory performance under the investigated high-speed train loading conditions and that no excessive deformations are expected to occur.

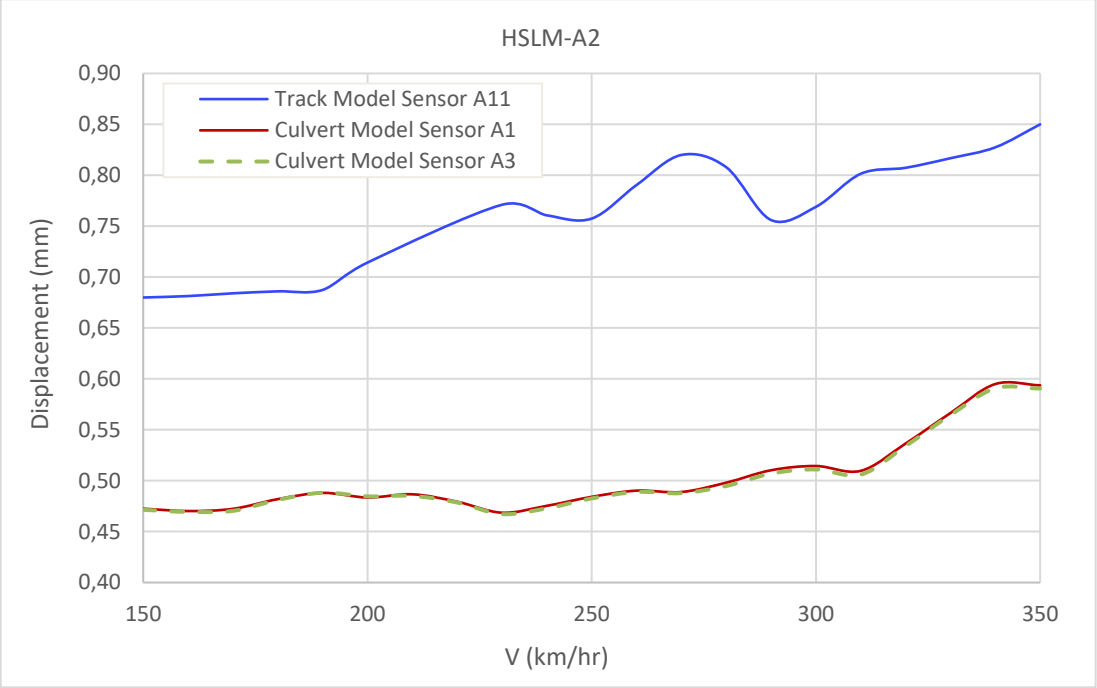


Figure 37: Maximum displacement responses at sensor A1 and A3 (culvert model) and A11 (track model) during the HSLM-A2 train design loads, filtered at 30 Hz.

The acceleration curves for all sensor locations show a generally increasing trend with increasing train speed, accompanied by minor local fluctuations. However, no abrupt increases in displacement are observed, which indicates the absence of resonance phenomena within the investigated speed range. To further assess the nature of these peaks, the corresponding acceleration responses were examined in the time domain at train speeds corresponding to the local maxima observed in the acceleration–speed curves. For the benchmark track model, the response was evaluated at 325 km/h, while for the calibrated culvert model the response was evaluated at 340 km/h. Representative time histories for the track and culvert models at the investigated speeds are shown in Figure 38.

As illustrated in the time-domain responses, the vibration amplitudes remain relatively stable throughout the train passage and do not show the progressive build-up of oscillations that are

characteristic of resonance conditions. Instead, the response consists of periodic load-induced vibrations associated with the passage of individual axle loads. The amplitudes increase and decrease as the train enters and leaves the structure, reflecting the transient nature of the excitation rather than a resonant amplification of the structural response. This observation confirms that the local peaks identified in the speed-response plots are related to variations in axle-loading frequencies and dynamic interaction effects rather than resonance phenomena.

The absence of resonance is also supported by comparison with the estimated Rayleigh wave velocity of the calibrated soil profile. The Rayleigh wave velocity of the subgrade soil was estimated to be approximately 435 $\frac{\text{km}}{\text{h}}$, which exceeds all train speeds considered in this study. Since the investigated train speeds remain below this critical wave propagation velocity, significant amplification associated with ground vibration resonance or critical velocity effects was not expected. Consequently, the results indicate that neither the track system nor the soil–steel culvert is likely to experience resonance or excessive dynamic amplification within the investigated high-speed operating range.

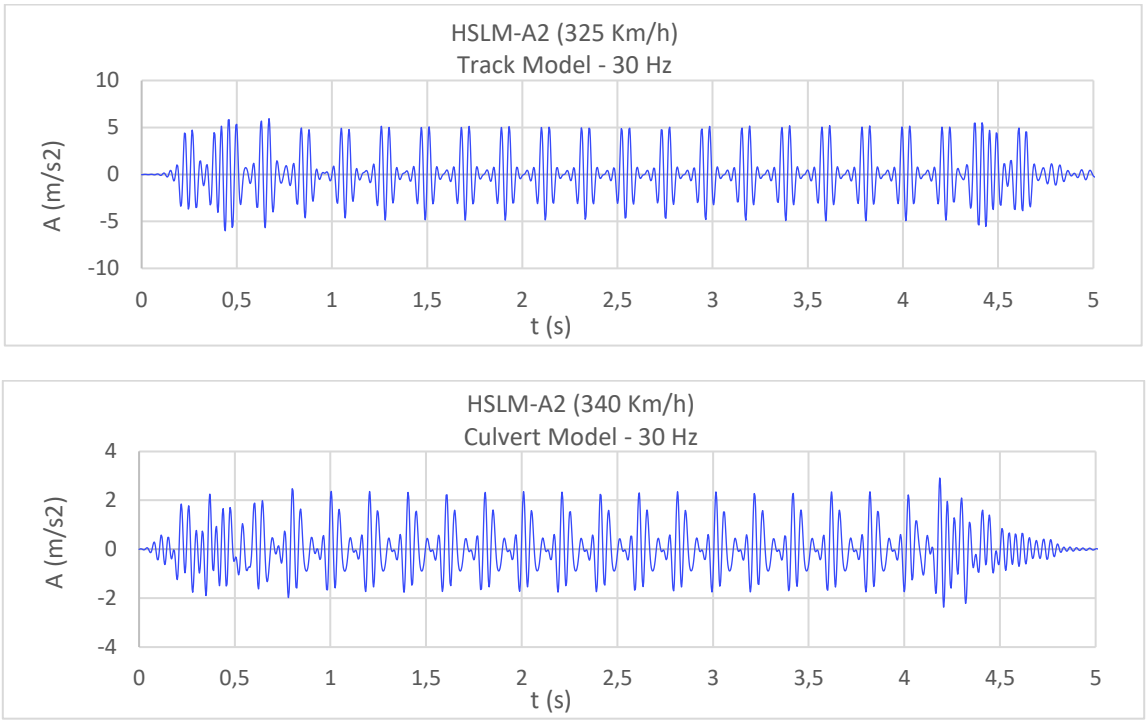


Figure 38: Acceleration time histories of the track and culvert models filtered at 30 Hz at sensor A1 and A11 for the HSLM-A2 train loading.

7 Conclusion and Future work

7.1 Conclusions

The field measurements showed that the adjacent track section and the soil–steel culvert exhibit similar fundamental vibration frequencies, with the first dominant mode occurring at approximately 19 Hz for the track section and 20 Hz for the culvert section. The measured train-passage responses also indicated that the adjacent track section experienced slightly higher acceleration amplitudes than the culvert midspan, suggesting that the culvert behaves as a relatively stiffer and more constrained system due to the beneficial soil–steel interaction.

Comparison between measured and simulated train-passage responses showed that the numerical models generally underestimated the response amplitudes by approximately a factor of two to three. This indicates that the numerical models are somewhat stiffer than the real structure. Nevertheless, the overall response patterns, dominant frequencies, and dynamic characteristics showed reasonable agreement with the field measurements, indicating that the models are suitable for evaluating the general dynamic behavior of the bridge and assessing potential resonance effects.

The high-speed train simulations showed that the adjacent track section produced higher acceleration and displacement responses than the culvert section throughout the investigated speed range. This large difference in amplitude is likely due to the overestimation of backfill soil stiffness in the culvert FE model.

No resonance behavior was observed within the investigated speed range. Although local response maxima were identified, neither the acceleration-speed curves nor the corresponding time-domain responses showed the characteristic amplification associated with resonance. Furthermore, the investigated train speeds remained below the estimated Rayleigh wave velocity of the calibrated soil profile, supporting the conclusion that critical velocity effects are unlikely to occur. However, these velocities were calculated from the calibrated soil properties used in the FE models, rather than from independent in-situ wave velocity measurements. Therefore, the calculated Rayleigh velocities should be interpreted only as approximate indicators supporting the Brigade analysis, rather than as definitive critical wave velocities.

For acceleration response, the Eurocode vertical deck acceleration limit of 3.5 $\frac{\mathrm{m}}{\mathrm{s}^2}$ for ballasted bridges can be used as a general reference value. Although some acceleration amplitudes exceeded this reference value, the time and frequency domain responses did not indicate resonance or progressive dynamic amplification within the investigated speed range. Therefore, the exceedance can be interpreted mainly as a local acceleration response rather than evidence of global dynamic instability.

Finally, even when applying a conservative correction factor of two to the simulated culvert displacements to account for the underestimation observed during validation, the maximum predicted displacement remained below the 2 mm serviceability limit specified by *Trafikverket*. Based on the loading conditions investigated, the Åre culvert is therefore expected to satisfy the displacement

requirements and is not anticipated to experience resonance or significant dynamic disturbances under high-speed train traffic.

7.2 Future Work

Although the calibrated numerical models provided satisfactory agreement with the measured responses, several opportunities exist for further improving the accuracy and reliability of the dynamic assessment. Some of the observed discrepancies in the results may be attributed to the simplified representation of the soil layers in the FE models. A more detailed modeling of the different soil layers could lead to better agreement between the numerical model results and measured field data.

Future studies can investigate the influence of seasonal variations on the dynamic behavior of soil–steel composite bridges. Soil stiffness and damping properties are known to vary with temperature, moisture content, and frost conditions, which may significantly influence the dynamic response of the structure. Repeating the field measurements during different seasons would therefore provide valuable information regarding the sensitivity of the bridge to environmental conditions and enable the development of more representative calibration parameters.

Finally, a future verification study performed on the same bridge using additional field measurements could be used to confirm the conclusions obtained in this research. Additional field measurements involving a larger number of train passages and train types would further improve the reliability of the validation process. More extensive monitoring could also help quantify the variability of the measured responses and reduce uncertainties associated with calibration.

Such measurements would provide valuable validation data and contribute to a better understanding of the long-term dynamic performance of soil–steel composite bridges subjected to increasing railway traffic demands.

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Appendix A: Mesh Convergence

A mesh convergence study was carried out for the developed finite element models to ensure that the obtained numerical responses were sufficiently independent of the selected mesh density. In finite element analysis, the geometry is discretized into smaller elements, and the accuracy of the solution is therefore influenced by the size and distribution of the mesh. A coarse mesh may lead to inaccurate representation of stiffness, wave propagation, and dynamic behavior, while an excessively refined mesh may significantly increase the computational cost without providing substantial improvements in accuracy. Consequently, a balance between numerical accuracy and computational efficiency must be achieved. The mesh convergence procedure was performed by progressively refining the mesh of the investigated model components and comparing the resulting dynamic responses. According to common finite element modelling practice, convergence is considered achieved when further refinement of the mesh produces only negligible changes in the calculated results.

In this case study, the convergence assessment was based on comparisons of the frequency response functions (FRFs) obtained for different mesh sizes at selected locations within the numerical models. Particular attention was given to the dominant resonance frequencies, response amplitudes, and the overall shape of the FRF curves within the frequency range of interest (1-30 Hz).

The convergence analyses were conducted separately for the soil domain, ballast layer in the track model, and steel culvert in the culvert model due to the different stiffness characteristics and expected wavelength propagation within each component. For each investigated mesh size, the numerical model was re-meshed and re-analyzed under identical loading and boundary conditions. The obtained FRFs were then compared to evaluate the influence of mesh refinement on the dynamic response. A mesh-independent solution was considered achieved when the response curves corresponding to consecutive mesh refinements showed only minor deviations. This process was performed before the transformation of mesh elements to infinite elements at the longitudinal boundaries of the model.

For the soil domain of the benchmark track model, the mesh convergence study was initiated using a relatively coarse mesh size of 1.0 m. The mesh was then progressively refined to element sizes of 0.8 m and 0.6 m, while monitoring the resulting frequency response functions (FRFs) at a node located on the soil part surface beneath the ballast at the track midspan, as shown in Figure 39. The comparison of the FRFs demonstrates that the overall response characteristics and dominant resonance frequencies remain relatively consistent as the mesh is refined. Only minor variations in the response amplitudes and higher-frequency behavior were observed between the investigated mesh densities. Since the response curves obtained for mesh sizes of 0.8 m and 0.6 m showed close agreement, the solution was considered sufficiently converged at a mesh size of 0.8 m. This mesh size was therefore selected for the soil domain in the final benchmark track model as it provided a satisfactory balance between computational efficiency and numerical accuracy.

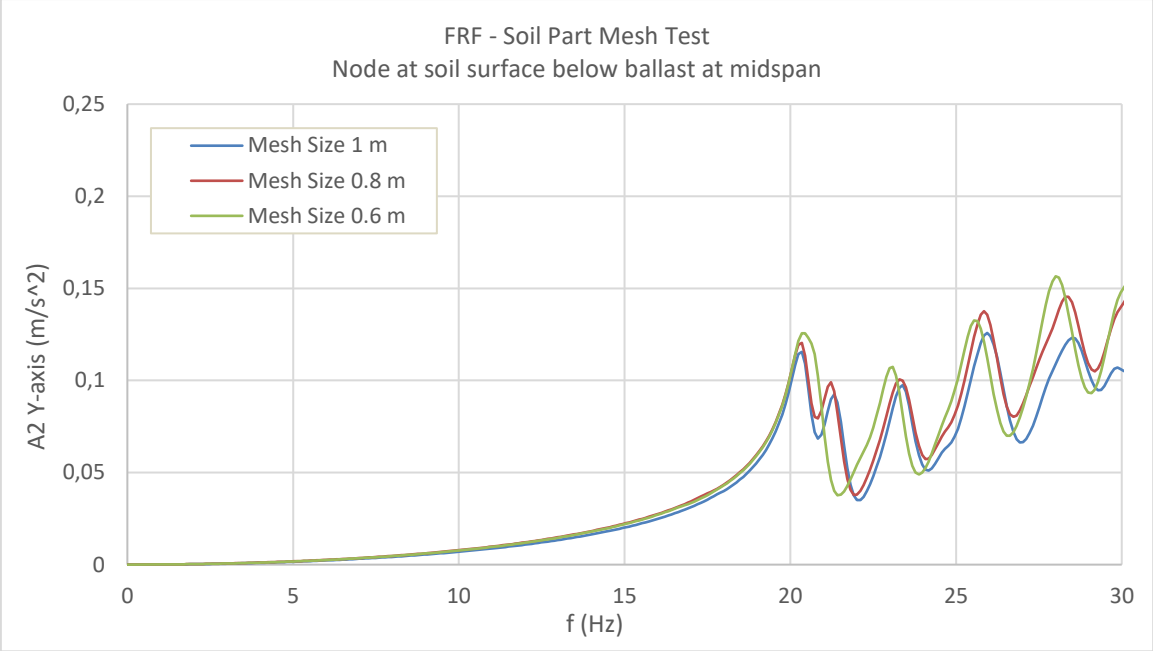


Figure 39: Mesh sensitivity analysis of the soil domain in the track model showing the frequency response functions (FRFs) obtained for different mesh sizes at a node located on the soil surface beneath the ballast at the track midspan.

A similar convergence procedure was performed for the ballast layer, where the initial mesh size was also defined as 1.0 m and subsequently refined to 0.8 m, 0.6 m, 0.4 m, 0.3 m, and 0.2 m. The corresponding FRFs were extracted at a node located on the ballast surface at the track midspan, as illustrated in Figure 40. Compared to the soil domain, the ballast layer exhibited greater sensitivity to mesh refinement due to its higher local stiffness variations and more pronounced dynamic response. As the mesh became finer, the FRF curves gradually stabilized, particularly with respect to the dominant resonance peaks and the overall response shape. The results indicate that the response obtained using mesh sizes of 0.3 m and 0.2 m showed only minor differences, indicating that mesh convergence had been achieved. Consequently, a mesh size of 0.3 m was adopted for the ballast layer in the final numerical model.

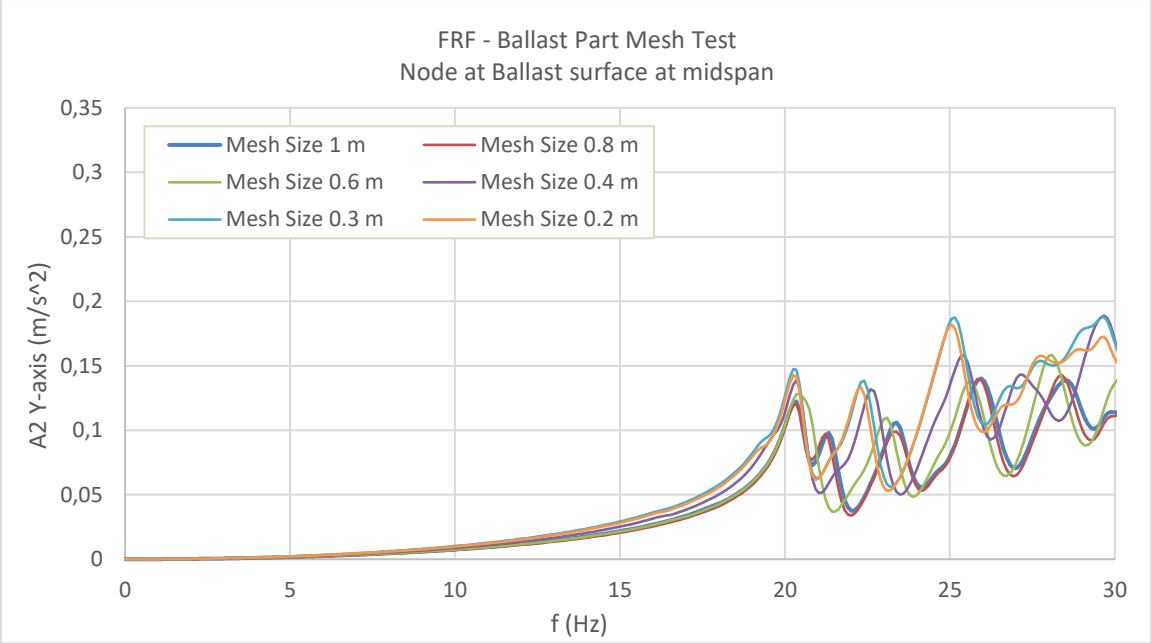


Figure 40: Mesh sensitivity analysis of the ballast layer in the track model showing the frequency response functions (FRFs) obtained for different mesh sizes at a node located on the ballast surface at the track midspan.

A mesh convergence study was also performed for the steel culvert component in the culvert model. Due to the relatively high stiffness of the corrugated steel structure and the expected local stress concentrations within the shell geometry, a finer mesh refinement was required compared to the surrounding soil domain. The convergence analysis was initiated using a mesh size of 0.6 m for the culvert shell elements, which were subsequently refined to 0.4 m and 0.3 m. The corresponding frequency response functions (FRFs) were evaluated at a node located on the top surface of the culvert at the culvert midspan, as presented in Figure 41.

The FRFs obtained demonstrate that the overall dynamic response, including the dominant resonance peaks and the general response behavior, remains relatively stable throughout the refinement process. Only minor differences can be observed between the investigated mesh sizes, particularly at higher frequencies. The response curves corresponding to mesh sizes of 0.4 m and 0.3 m show close agreement, indicating that further mesh refinement produces only limited changes in the numerical response. Based on the convergence results and considering the computational efficiency of the model, a mesh size of 0.4 m was adopted for the steel culvert in the final calibrated culvert model.

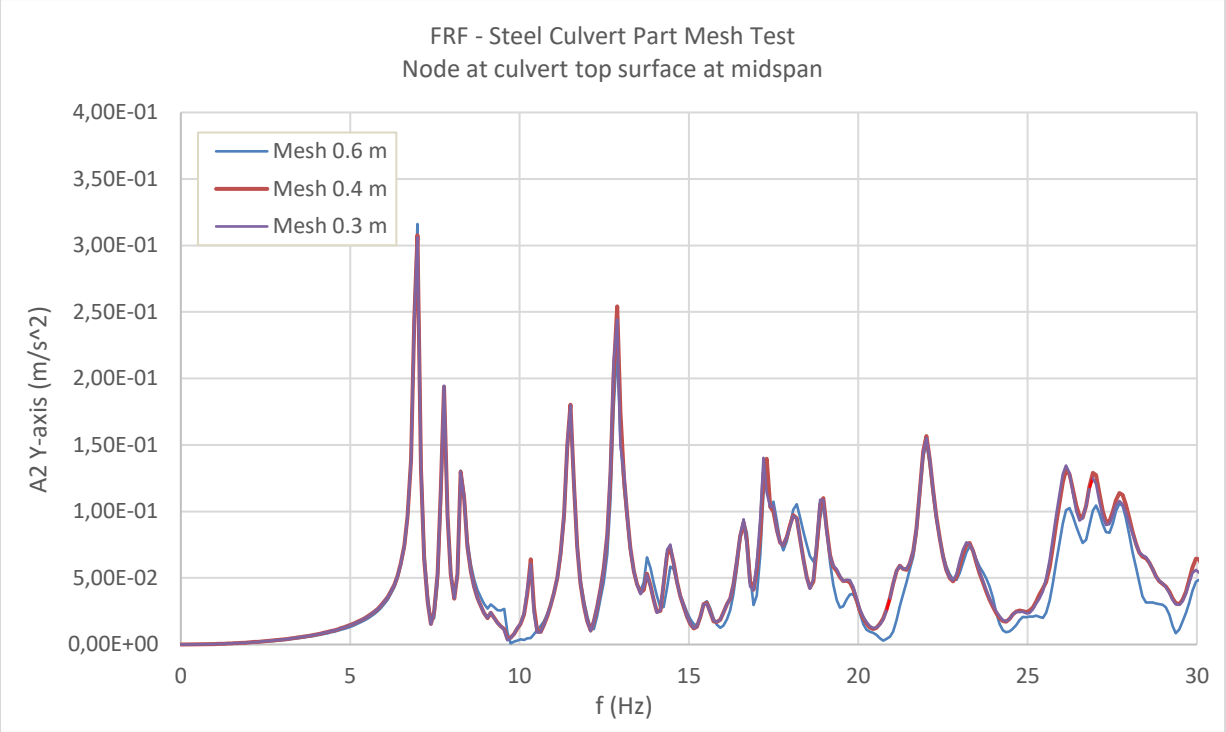


Figure 41: Mesh sensitivity analysis of the steel culvert in the culvert model showing the frequency response functions (FRFs) obtained for different mesh sizes at a node located on the top surface of the culvert at midspan.

Appendix B: C3D to CIN3D FEM Element Conversion

```
function C3D_to_CIN3D(name_Inflayer)
% Convert Abaqus/BrigADE Input Model C3D10/C3D10R/C3D10RH elements to CIN3D12R
%   C3D_to_CIN3D('Inflayer')

    if nargin < 1 || isempty(name_Inflayer)
        error('You must provide a part name, e.g. C3D_to_CIN3D(''Inflayer'')');
    end

    % Select input file
    [file, dr] = uigetfile('*.inp', 'Select Abaqus/BrigADE *.inp file');
    if isequal(file,0)
        disp('Cancelled.');
        return;
    end
    inPath = fullfile(dr, file);

    % Read all lines
    L = readlines(inPath);
    L = erase(L, char(13));

    % Find part
    partPattern = ['^\*Part\s*,\s*name\s*=\s*' regexptranslate('escape', name_Inflayer) '\s*$'];
    idxPartStart = find(~cellfun('isempty', regexp(cellstr(L), partPattern, 'ignorecase')), 1,
'first');
    if isempty(idxPartStart)
        error('Could not find *Part, name=%s in the file.', name_inf);
    end

    % Find end part. Mesh element stack direction assigned in Brigade
    idxPartEnd = [];
    for k = idxPartStart+1:numel(L)
        if startsWith(strtrim(L(k)), '*End Part', 'IgnoreCase', true)
            idxPartEnd = k;
            break;
        end
    end
    if isempty(idxPartEnd)
        error('Could not find *End Part for part %s.', name_inf);
    end

    % Find first *Element block inside Infinte Layer part
    idxElemHdr = [];
    elemType = "";
    for k = idxPartStart:idxPartEnd
        s = strtrim(L(k));
        if startsWith(s, '*Element', 'IgnoreCase', true)
            idxElemHdr = k;
            tok = regexp(char(s), 'type\s*=\s*([^,\s]+)', 'tokens', 'once', 'ignorecase');
            if ~isempty(tok)
                elemType = upper(string(tok{1}));
            end
            break;
        end
    end

    if isempty(idxElemHdr)
        error('No *Element block found inside part %s.', name_inf);
    end

    if ~(elemType == "C3D20" || elemType == "C3D20R" || elemType == "C3D20RH")
```

```
60          error('Element block in part %s is type %s. Only C3D20/C3D20R/C3D20RH are supported.',
    name_Inflayer, elemType);
61      end
62
63      % Read element lines until next keyword
64      elemLines = strings(0,1);
65      idxAfterBlock = idxElemHdr + 1;
66      for k = idxElemHdr+1:idxPartEnd
67          s = strtrim(L(k));
68          if startsWith(s, '*')
69              idxAfterBlock = k;
70              break;
71          end
72          if strlength(s) > 0
73              elemLines(end+1,1) = L(k);
74          end
75      end
76
77      % Parse eid + 20 node ids
78      EID = [];
79      N20 = [];
80      buf = [];
81
82      for k = 1:numel(elemLines)
83          sNum = replace(elemLines(k), ',', ' ');
84          vals = sscanf(sNum, '%f').';
85          if isempty(vals)
86              continue;
87          end
88
89          buf = [buf, vals];
90
91          while numel(buf) >= 21
92              eid = buf(1);
93              nodes20 = buf(2:21);
94              EID(end+1,1) = eid;
95              N20(end+1,:) = nodes20;
96              buf = buf(22:end);
97          end
98      end
99
100     if isempty(N20)
101         error('No C3D20 elements could be parsed inside part %s.', name_inf);
102     end
103
104     % Remap elements C3D20 -> CIN3D12R
105     N12 = N20(:, [1:4 9:12 5:8]);
106
107     % Write output
108     [~, base, ~] = fileparts(file);
109     outPath = fullfile(dr, sprintf('%s_CIN3D12R.inp', base));
110
111     fid = fopen(outPath, 'w');
112     if fid < 0
113         error('Could not open output file for writing: %s', outPath);
114     end
115
116     for k = 1:idxElemHdr-1
117         fprintf(fid, '%s\n', L(k));
118     end
119
```

```
120     fprintf(fid, '*ELEMENT, TYPE=CIN3D12R\n');
121
122     for i = 1:size(N12,1)
123         fprintf(fid, '%.0f, %.0f, %.0f, %.0f, %.0f, %.0f, %.0f, %.0f, %.0f, %.0f, %.0f, %.0f,
    %.0f\n', ...
124             EID(i), N12(i,:));
125     end
126
127     for k = idxAfterBlock:numel(L)
128         fprintf(fid, '%s\n', L(k));
129     end
130
131     fclose(fid);
132
133     fprintf('\nDONE.\n');
134     fprintf('Input : %s\n', inPath);
135     fprintf('Output: %s\n', outPath);
136     fprintf('Converted %d elements in part %s to CIN3D12R.\n', size(N12,1), name_Inflayer);
137 end
```